Supplementary Material for: Velocity-dependent inertial induction in spiral galaxies: a screened Proca current–interaction framework and SPARC audit

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S1 SUPPLEMENTARY MATERIAL

This supplementary file accompanies the main manuscript and contains extended derivations, additional robustness checks, the detailed resolved-map protocol, and the full reproducibility tables referred to in the main text.

S1 PROCA–GEM ACTION, RELATION TO WEAK-FIELD GRAVITY, AND QUASI-STATIC REDUCTION

S1.1 Covariant definition of the screened current sector

A minimal covariant definition of the screened current sector is given by a massive vector (Proca) field A_μ universally coupled to a conserved mass–energy current J^μ,

$$S = \int d^4x \sqrt{-g}\left[\frac{1}{16\pi G_0} R - \frac{1}{4} F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\kappa^2 A_\mu A^\mu - \frac{4\pi G_0}{c} A_\mu J^\mu\right], \tag{S1}$$

with $F_{\mu\nu} \equiv \nabla_\mu A_\nu - \nabla_\nu A_\mu$. The minus sign in the source term of the action is chosen so that variation with respect to A_μ yields the source convention used in equation (S2). The Proca mass term breaks gauge invariance at finite κ but the massless limit $\kappa \to 0$ restores the gauge symmetry and reduces the current sector to a Maxwell-like GEM form. Universal coupling is consistent at the level of the equations of motion provided $\nabla_\mu J^\mu = 0$ (the current conservation assumed throughout).

S1.2 Field equations and quasi-static Helmholtz reduction

Varying (S1) with respect to A_μ yields the Proca equation with source,

$$\nabla_\nu F^{\mu\nu} + \kappa^2 A^\mu = \frac{4\pi G_0}{c} J^\mu, \tag{S2}$$

and taking the covariant divergence gives the Lorenz-type constraint $\kappa^2 \nabla_\mu A^\mu = (4\pi G_0/c)\nabla_\mu J^\mu$, so that $\nabla_\mu A^\mu = 0$ follows when $\nabla_\mu J^\mu = 0$. In the weak-field, slow-motion, quasi-static regime, the field equations reduce to the screened Poisson/Helmholtz system of Eq. (**??**), with $J^\mu \simeq (c\rho, \rho\mathbf{v})$ and $\nabla\cdot\mathbf{A} \simeq 0$.

Source-sign convention. In the quasi-static reduction to the Helmholtz system (Eq. (**??**)), the gravitational convention $\nabla^2\mathbf{A} - \kappa^2\mathbf{A} = +(4\pi G_0/c)\,\rho\mathbf{v}$ is adopted, following the standard GEM practice in which the overall coupling sign is chosen so that the gravitomagnetic interaction is attractive for co-rotating mass currents. This differs from the electromagnetic sign by the usual gravitational-vs-electromagnetic flip and corresponds to absorbing a factor of (-1) into the definition of the GEM potentials relative to the covariant A_μ field in equation (S2). All main-text equations, the Green-function solution, and the annulus operator use this gravitational convention consistently.

S1.3 Relationship to the weak-field GEM limit

The potentials $(\Phi, \mathbf{A})$ enter the slow-motion force law through the standard Lorentz-form GEM channel, with the induced contribution arising from $\mathbf{v} \times \mathbf{B}_g$ where $\mathbf{B}_g \equiv \nabla \times \mathbf{A}$. The conservative quasi-static sector employed for rotation-curve closure uses only this time-symmetric interaction channel. Any time-asymmetric completion (radiation-reaction / background dissipation) is treated as a separate modelling layer and is not required for the quasi-static force balance used in the SPARC fits.

S1.4 Screening scale and external-constraint map

The screening wavenumber κ is calibrated by the cosmological matching condition adopted in the parent theory and is held fixed in the SPARC analysis. For internal consistency and scope clarity, the following constraint channels should be checked against the calibrated κ: (i) Solar-System Yukawa/PPN bounds on long-range additional fields; (ii) binary-pulsar timing constraints on additional propagating degrees of freedom; (iii) gravitational-wave propagation constraints (e.g. dispersion) relevant to extremely light fields; (iv) cosmological background/structure constraints if the field carries non-negligible energy density. This manuscript restricts attention to the galactic near-zone regime where $\kappa R \ll 1$ over SPARC radii, while retaining κ as a fixed theory parameter in the parent equations.

S1.5 On theoretical naturalness and parameter regime

The Proca mass scale corresponding to κ is treated as a phenomenological screening parameter calibrated by cosmological matching in the parent theory. From an effective-field-theory perspective, the extremely light-mass regime is technically natural only if protected by a symmetry in the $\kappa \to 0$ limit (restored gauge symmetry) and if radiative corrections are controlled by the assumed ultraviolet completion. A complete assessment of naturalness and of all fifth-force / PPN / pulsar / gravitational-wave propagation constraints is deferred to the Solar-System and compact-object programme papers of the series; the present manuscript restricts attention to the galactic near-zone regime and provides a transparent map of the relevant constraint channels.

S2 DERIVATION OF THE RAZOR-THIN ANNULUS OPERATOR AND PRINCIPAL-VALUE KERNEL

S2.1 Midplane projection and azimuthal current source

Assume a razor-thin, axisymmetric disc with surface density $\Sigma(R)$ and circular streaming $v_\phi(R)$ so that the mass current is

$$\mathbf{J}(\mathbf{x}) \;=\; \Sigma(R)\, v_\phi(R)\, \delta(z)\, \hat{\boldsymbol{\phi}}, \qquad J_\phi(R,z) \;=\; \Sigma(R)\, v_\phi(R)\, \delta(z). \tag{S3}$$

In the quasi-static limit, the ϕ component of the screened vector potential satisfies

$$(\nabla^2 - \kappa^2) A_\phi(\mathbf{x}) \;=\; \frac{4\pi G_0}{c}\, J_\phi(\mathbf{x}). \tag{S4}$$

S2.2 Green-function solution and azimuthal integration

Using the 3D Helmholtz Green function $G_\kappa(\mathbf{x},\mathbf{x}') = \exp(-\kappa|\mathbf{x}-\mathbf{x}'|)/|\mathbf{x}-\mathbf{x}'|$, the midplane solution is

$$\begin{aligned} A_\phi(R,0) \;&=\; \frac{4\pi G_0}{c} \int_0^\infty dR'\, R'\, \Sigma(R') v_\phi(R') \\ &\quad\times \int_0^{2\pi} \frac{d\phi'}{2\pi}\, \frac{e^{-\kappa\Delta}}{\Delta}\, \cos(\phi-\phi'), \end{aligned} \tag{S5}$$

where $\Delta = \sqrt{R^2 + R'^2 - 2RR'\cos(\phi-\phi')}$ and the cosine factor is the projection onto the $\hat{\boldsymbol{\phi}}$ direction for an axisymmetric source.

S2.3 From A_ϕ to a radial acceleration

For circular motion $\mathbf{v} = v_\phi(R)\hat{\boldsymbol{\phi}}$, the conservative induced contribution in the midplane follows from the Lorentz channel $\mathbf{a}_{\rm ind} \propto (\mathbf{v}/c)\times\mathbf{B}_g$ with $\mathbf{B}_g = \nabla\times\mathbf{A}$. In axisymmetry with dominant $A_\phi(R,0)$, the relevant component is

$$\begin{aligned} B_{g,z}(R,0) &= \frac{1}{R}\frac{\partial}{\partial R}\big(R A_\phi(R,0)\big), \\ g_{\rm ind,bare}(R) &\propto \frac{v_\phi(R)}{c}\, B_{g,z}(R,0), \end{aligned} \tag{S6}$$

so the induced term is a disc-integrated functional of the azimuthal current Σv_ϕ.

S2.4 Origin of the $(R^2 - R'^2)^{-1}$ structure and PV prescription

After azimuthal integration, the midplane kernel develops a removable singularity at co-rotation $R' = R$ originating from the local Δ^{-1} behaviour of the Green function and the derivative in Eq. (S6). In the razor-thin limit this appears as a rational factor proportional to $(R^2 - R'^2)^{-1}$ multiplying a signed-square velocity combination that encodes the directionality of the differential rotation. The disc is therefore naturally decomposed into interior and exterior contributions,

$$g_{\rm ind,bare}(R) \;=\; C_\nu \Big[P_{\rm bare}(R) - D_{\rm bare}(R)\Big], \tag{S7}$$

with $P_{\rm bare}$ integrating over $R' < R$ (interior annuli) and $D_{\rm bare}$ integrating over $R' > R$ (exterior annuli). The removable co-rotation singularity is evaluated as a regulated Cauchy principal value,

$$(R^2 - R'^2)^{-1} \;\longrightarrow\; (R^2 - R'^2 + \varepsilon^2 R^2)^{-1}, \tag{S8}$$

and ε is verified by the principal-value regulator robustness sweep reported in Table S1 to lie in the converged regime for the adopted configuration.

S3 QUASI-STATIC CURRENT SECTOR AND THIN-DISC REDUCTION

S3.1 Action-level definition and weak-field correspondence

The screened Proca–GEM sector used throughout is defined by the covariant action (Paper I)

$$S = \int d^4x\, \sqrt{-g}\, \left[-\frac{c^4}{16\pi G_0}\Big(F^{\mu\nu}F_{\mu\nu} + 2\kappa^2 A^\mu A_\mu\Big) - \frac{1}{c} A_\mu J'^\mu \right], \tag{S9}$$

with $F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$ and conserved source current $J'^\mu = (c\rho, \rho\mathbf{v})$ in the weak-field limit. Variation with respect to A_μ yields the Proca field equations and Lorenz condition,

$$\partial_\nu F^{\mu\nu} - \kappa^2 A^\mu = \frac{4\pi G_0}{c^3} J'^\mu, \qquad \partial_\mu A^\mu = 0, \tag{S10}$$

which reduce, in the quasi-static slow-motion regime, to the Helmholtz system written in Eq. (**??**). Paper I further shows that the near-zone post-Newtonian mapping reproduces the standard GR values of the Eddington–Robertson PPN parameters up to screening-suppressed corrections, $\gamma = \beta = 1 + O((\kappa r)^2)$, ensuring compatibility with Solar-System bounds in the $\kappa r \ll 1$ regime.
Convention note. The field normalisation adopted in this appendix absorbs a factor of $c^2/\sqrt{4\pi G_0}$ relative to the canonically normalised Proca field used in Appendix S1. Under this rescaling, the sign of the mass term and the source coupling constant change in appearance but the physical content—including the quasi-static Helmholtz reduction in Eq. (**??**) and the Yukawa near-zone behaviour—is identical. All main-text equations use the Appendix S1 convention throughout.

S3.2 Helmholtz structure and near-zone limit

The screened Proca–GEM potentials satisfy the Helmholtz-type equations (Eq. (**??**)). In the galactic near-zone regime relevant for SPARC radii, $\kappa R \ll 1$ holds while κ remains fixed by the cosmological matching condition of the parent theory. Expanding the Yukawa Green function gives

$$\frac{e^{-\kappa r}}{r} = \frac{1}{r}\left[1 - \kappa r + \frac{1}{2}(\kappa r)^2 + O((\kappa r)^3)\right], \tag{S11}$$

while the corresponding static force carries the combination

$$e^{-\kappa r}(1 + \kappa r) = 1 - \frac{1}{2}(\kappa r)^2 + O((\kappa r)^3). \tag{S12}$$

Thus the leading fractional correction to the Newtonian $1/r^2$ force is quadratic in κr, so exponential attenuation is negligible across disc scales even though the screening parameter remains part of the theory-level calibration.

The conservative quasi-static sector retained in the rotation-curve audit therefore corresponds to the leading-order Helmholtz solution with fixed κ and fixed normalisation $C_v = C_\star G_0/c^2$.

At galactic radii the application indeed lies in the near-zone regime $\kappa R \ll 1$, so exponential Yukawa attenuation is negligible across the disc. The role of the Proca embedding is therefore not to generate a directly observable galaxy-scale suppression factor, but to supply a covariant screened current sector with a controlled weak-field limit, a well-defined massless limit, and a consistent bridge to the broader matching conditions used elsewhere in the programme. In the SPARC application itself, the observable phenomenology is governed by the nonlocal current–current annulus operator rather than by local Yukawa damping.

S3.3 Principal-value annulus operator

In the razor-thin, axisymmetric disc limit with surface density $\Sigma(R)$ and azimuthal velocity $v_\phi(R)$, the current sector reduces to a one-dimensional annulus operator acting on the baryonic surface mass current. The induced radial acceleration is written as

$$g_{\rm ind,bare}(R) = C_v\Big[P_{\rm bare}(R) - D_{\rm bare}(R)\Big], \tag{S13}$$

where $P_{\rm bare}$ and $D_{\rm bare}$ denote inner and outer annulus contributions, respectively.

The kernel contains a removable singularity at co-rotation $R' = R$. This is treated using a regulated Cauchy principal-value prescription,

$$(R^2 - R'^2)^{-1} \;\rightarrow\; (R^2 - R'^2 + \varepsilon^2 R^2)^{-1}, \tag{S14}$$

with ε fixed under the adopted SPARC policy and convergence verified explicitly (Fig. **??**).

S3.4 From Helmholtz solution to the principal-value annulus kernel (sketch)

In an axisymmetric razor-thin disc the only sourced component of the vector potential is the azimuthal component $A_\phi(R, z)$, because the surface mass current is dominantly azimuthal: $J_\phi(R) \simeq \Sigma(R)\, v_\phi(R)\, \delta(z)$. The quasi-static Helmholtz equation for A_ϕ therefore admits a Green-function solution of the form $A_\phi(R, 0) = \int dR'\, R'\, J_\phi(R')\, \mathcal{G}(R, R'; \kappa)$ after azimuthal integration. The relevant gravito-magnetic field component entering the radial Lorentz channel is $B_{g,z}(R, 0) = (1/R)\, \partial_R\big(RA_\phi(R, 0)\big)$, so that the induced radial acceleration takes the schematic form $g_{\rm ind,bare}(R) \propto (v_\phi/c)\, B_{g,z}(R, 0)$. Differentiating the azimuthally integrated Green-function kernel generates the rational factor $(R^2 - R'^2)^{-1}$; the singularity at $R' = R$ is removable and is implemented as a Cauchy principal value, leading to the regulated prescription in Eq. (S14). The resulting pull/drag split arises from separating the integration domain into interior ($R' < R$) and exterior ($R' > R$) annuli, yielding $P_{\rm bare}$ and $D_{\rm bare}$, respectively.

S3.5 Two-body matching and kernel normalisation

The overall normalisation $C_v = C_\star G_0/c^2$ is fixed by matching the annulus reduction to the velocity-dependent two-body interaction in the weak-field limit. This fixes $C_\star$ uniquely and removes any free normalisation from the galaxy analysis. No re-fitting of C_v is performed at catalogue or galaxy level. All rotation-curve closures therefore test the thin-disc operator with theory-level calibration only.

SUPPLEMENTARY APPENDICES RETAINED IN THE SUBMISSION DRAFT

The appendices collected below provide supporting derivations, numerical robustness checks, reproducibility details, extended comparison diagnostics, and a map-level validation protocol. They are retained in the present submission draft for completeness and ease of review, but can be detached as online supplementary material if the journal page limit requires it.

S4 ANNULUS KERNEL, REGULATOR ROBUSTNESS, AND SPARC CUTS

S4.1 Dimensionless annulus kernel representation

Let the disc be truncated at radius $R_{\max}$ and define dimensionless variables

$$\xi \equiv \frac{R'}{R_{\max}}, \qquad \eta \equiv \frac{R}{R_{\max}}, \qquad 0 \le \xi, \eta \le 1. \tag{S15}$$

Let $\Sigma(\xi)$ denote the baryonic surface density and $v(\xi)$ the azimuthal speed under the adopted mass-model convention. After azimuthal integration in the quasi-static thin-disc limit, the inner (pull) and outer (drag) annulus columns may be written schematically as

$$P_{\rm bare}(\eta) = \frac{8}{3}\int_0^{\eta} \mathcal{K}_-(\eta,\xi)\,\Sigma(\xi)\,\mathcal{J}_-(\eta,\xi)\,\mathrm{d}\xi, \tag{S16}$$

$$D_{\rm bare}(\eta) = \frac{8}{3}\int_{\eta}^{1} \mathcal{K}_+(\eta,\xi)\,\Sigma(\xi)\,\mathcal{J}_+(\eta,\xi)\,\mathrm{d}\xi. \tag{S17}$$

where the geometric kernels are

$$\mathcal{K}_-(\eta,\xi) = \frac{\xi^2}{\eta^3(\eta^2-\xi^2)}, \tag{S18}$$

$$\mathcal{K}_+(\eta,\xi) = \frac{1}{\xi(\xi^2-\eta^2)}, \tag{S19}$$

and the current bilinears are

$$\mathcal{J}_-(\eta,\xi) = \left[\eta v(\xi) - \xi v(\eta)\right]\left|\eta v(\xi) - \xi v(\eta)\right|, \tag{S20}$$

$$\mathcal{J}_+(\eta,\xi) = \left[\xi v(\eta) - \eta v(\xi)\right]\left|\xi v(\eta) - \eta v(\xi)\right|. \tag{S21}$$

The full induced acceleration is then

$$g_{\rm ind,bare}(R) = C_v\left[P_{\rm bare}(\eta) - D_{\rm bare}(\eta)\right], \tag{S22}$$

with $C_v = C_\star G_0/c^2$ fixed by two-body matching. These expressions make explicit that the operator is bilinear in the surface mass current and contains no adjustable radial functions.

S4.2 Sign structure of $P_{\rm bare} - D_{\rm bare}$

The razor-thin axisymmetric reduction expresses the bare induced term as a difference of an interior (pull) and exterior (drag) contribution, $g_{\rm ind,bare}(R) = C_v\,[P_{\rm bare}(R) - D_{\rm bare}(R)]$. In the dimensionless annulus form (Appendix S4.1), both $P_{\rm bare}$ and $D_{\rm bare}$ can be written as positive-weight integrals over the same signed-square streaming mismatch

$$\Delta(\xi,\eta) \;\equiv\; \eta\, v(\xi) - \xi\, v(\eta), \qquad \Delta|\Delta| = \Delta^2\,\mathrm{sgn}(\Delta),$$

with $\eta \equiv R/R_{\max}$ and $\xi \equiv R'/R_{\max}$, where $R_{\max}$ is taken as the outermost SPARC radius with a valid rotation-curve measurement for each galaxy (i.e., the largest R_i passing the quality cuts of Appendix S4.4).

Dimensional consistency. In equations (S16)–(S17), the annulus integrals are written in dimensionless radii (η,ξ) with $R = \eta R_{\max}$ and $R' = \xi R_{\max}$, so that the physical measure obeys $\mathrm{d}R' = R_{\max}\,\mathrm{d}\xi$. Accordingly, while the geometric kernel factors are dimensionless functions of (η,ξ), the operator columns inherit dimensions from the surface density and velocity factors together with the $R_{\max}$ measure. With Σ in units of mass per area and v in units of length per time, the bare operator columns satisfy

$$[P_{\rm bare}] = [D_{\rm bare}] = [R_{\max}]\,[\Sigma]\,[v]^2 = \mathrm{mass\,time}^{-2},$$

Table S1. Principal-value regulator robustness test. Representative ε sweep for NGC 3198 under the fixed-normalisation kernel mode. $\Delta\chi^2$ is measured relative to $\varepsilon = 10^{-3}$.

ε	Max. fractional change in $g_{\rm ind}$	$\Delta\chi^2$
10^{-2}	$< 2\%$	< 0.5
5×10^{-3}	$< 1\%$	< 0.2
10^{-3}	reference	0
5×10^{-4}	$< 0.5\%$	< 0.1

and the prefactor $C_\nu = C_\star G_0/c^2$ carries dimensions $[C_\nu] = \text{length}\,\text{mass}^{-1}$. Therefore $g_{\rm ind,bare}(R) = C_\nu\,[P_{\rm bare}(R) - D_{\rm bare}(R)]$ has the correct dimensions of acceleration, $[g_{\rm ind,bare}] = \text{length}\,\text{time}^{-2}$. The accompanying operator-export metadata records the unit conventions (for Σ, v, and $R_{\rm max}$) used to construct the unit-calibrated columns. Schematically,

$$\begin{aligned} P_{\rm bare}(\eta) &= \int_0^\eta d\xi\; W_P(\eta,\xi)\,\Sigma(\xi)\,\Delta(\xi,\eta)^2, \\ D_{\rm bare}(\eta) &= \int_\eta^1 d\xi\; W_D(\eta,\xi)\,\Sigma(\xi)\,\Delta(\xi,\eta)^2, \end{aligned} \tag{S23}$$

where W_P and W_D are non-negative geometric weights determined by the thin-disc kernel (including the PV regulator). The sign condition required for additional centripetal support is therefore the explicit balance inequality

$$P_{\rm bare}(\eta) > D_{\rm bare}(\eta) \quad\Longleftrightarrow\quad \int_0^\eta d\xi\; W_P\,\Sigma\,\Delta^2 \;>\; \int_\eta^1 d\xi\; W_D\,\Sigma\,\Delta^2. \tag{S24}$$

Most discs satisfy decreasing angular speed $\Omega(R) = v_\phi(R)/R$ over the radii where rotation curves are measured, equivalently $\mathrm{d}(Rv_\phi)/\mathrm{d}R > 0$, which ensures that interior annuli rotate faster than the test radius while exterior annuli rotate more slowly. This establishes the physical pull/drag split but does not, by itself, enforce the inequality (S24): the outcome depends on the kernel-weighted competition between the centrally concentrated interior surface-mass current and the exterior contribution, and can be altered by steep outer declines in v_ϕ and/or sharp truncations in $\Sigma(R)$ near $R_{\rm max}$. The sign structure is therefore treated as an empirical diagnostic of the axisymmetric truncation; Fig. **??** provides a unit-calibrated example in which $P_{\rm bare} > D_{\rm bare}$ across the SPARC radii for NGC 3198.

S4.3 Principal-value regulator and ε sweep

The removable singularity at $\xi = \eta$ is treated using the regulated principal-value prescription

$$(\eta^2 - \xi^2)^{-1} \;\to\; (\eta^2 - \xi^2 + \varepsilon^2\eta^2)^{-1}. \tag{S25}$$

To verify that outer-disc behaviour is not a numerical artefact, a regulator sweep was performed over

$$\varepsilon \in \{10^{-2},\, 5\times10^{-3},\, 10^{-3},\, 5\times10^{-4}\}.$$

For representative galaxies (e.g. NGC 3198), the induced acceleration profile and total χ^2 were recomputed at each value. Results are summarised in Table S1.

The induced contribution is therefore numerically stable for $\varepsilon \lesssim 10^{-3}$, consistent with the behaviour illustrated in Fig. **??**.

S4.4 SPARC data cuts and quality masks

All catalogue-level diagnostics employ uniform data cuts applied identically across galaxies:

- Removal of radii flagged in the SPARC catalogue as unreliable or beam-smeared beyond quoted uncertainties.
- Exclusion of galaxies with inclination $i < 30^\circ$ to suppress large deprojection corrections.
- Adoption of SPARC quality flags $Q \le 2$.
- Minimum radial sampling threshold of $N_{\rm pts} \ge 5$.

The effective uncertainty is defined by Eq. (**??**) with fixed intrinsic astrophysical scatter parameter 0.2 = 0.2.

No per-galaxy distance or inclination re-fitting is performed in the RAR-surrogate configuration. When comparisons are drawn with alternative frameworks, matched cuts and identical uncertainty definitions are enforced to ensure that AIC/BIC penalties reflect parameter-count differences rather than data-selection effects.

S5 TWO-BODY MAPPING, SCALING ESTIMATES, AND QUASI-STATIC LIMITATIONS

S5.1 Mapping of the annulus operator to the two-body limit

The thin-disc pull–drag operator is constructed as the continuum (annulus) limit of a pairwise current interaction in the quasi-static Proca–GEM sector. Consider two point masses m and m' separated by $\mathbf{r}$ with in-plane velocities $\mathbf{v}$ and $\mathbf{v}'$. In the screened Proca–GEM quasi-static limit, the vector potential sourced by the moving mass has the Yukawa–Helmholtz form

$$\mathbf{A}(\mathbf{r}) \;=\; \frac{G_0 m'}{c}\,\frac{e^{-\kappa r}}{r}\,\mathbf{v}', \tag{S26}$$

so that the associated gravitomagnetic field is $\mathbf{B}_g = \nabla\times\mathbf{A}$. The conservative Lorentz-type channel then yields an acceleration of the schematic form

$$\mathbf{a}_{\rm ind} \;\propto\; \frac{G_0 m'}{c^2}\,\frac{e^{-\kappa r}}{r^2}\,\mathbf{v}\times(\mathbf{v}'\times\hat{\mathbf{r}}), \tag{S27}$$

whose in-plane projection produces the signed bilinear dependence on relative streaming that underlies the annulus reduction.

In the disc application, m' is replaced by an infinitesimal mass element $dm' = \Sigma(R')\,R'\,dR'\,d\theta$ on a circular orbit, and the azimuthal projection is integrated over θ. The resulting one-dimensional operator splits naturally into interior (pull) and exterior (drag) contributions because the sign of the relative angular streaming reverses across co-rotation. After the azimuthal integration, the two-body signed-square structure reduces to the dimensionless bilinear forms

$$\mathcal{J}_{\mp}(\eta,\xi) = \big[\eta v(\xi) - \xi v(\eta)\big]\big|\eta v(\xi) - \xi v(\eta)\big|$$

used in Appendix S4, with the geometric kernels $\mathcal{K}_{\mp}(\eta,\xi)$ determined uniquely by the thin-disc geometry.

The overall normalisation of the annulus operator is written as

$$C_v \equiv C_\star\,\frac{G_0}{c^2}, \tag{S28}$$

and the dimensionless coefficient $C_\star$ is fixed by matching the annulus limit to the same two-body interaction assumed in the Sciama–Ghosh inertial-induction development. Operationally, $C_\star$ is the angular-average coefficient produced by the θ-integration of the projected two-body Lorentz channel and is therefore not available for tuning in the galaxy analysis. The angular integration over the two-body Lorentz channel (Appendix S3.5) yields the fixed matching constant $C_\star = 8/3$. With $= 6.67430\times10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$ and $c = 2.99792458\times10^{8}\,\mathrm{m\,s^{-1}}$, the kernel normalisation is

$$C_v \;\equiv\; C_\star\,\frac{}{c^2} \;=\; 1.98\times10^{-27}\,\mathrm{m\,kg^{-1}}. \tag{S29}$$

With this choice, the thin-disc operator contains no adjustable coupling beyond the separately modelled bounded susceptibility $\mathcal{A}_{\rm eff}(R)$.

The apparent singularity at co-rotation ($R' = R$) corresponds in the two-body picture to vanishing relative streaming and is removable: the signed-square bilinear tends to zero as $\Delta v_\phi \to 0$ while the geometric factor diverges only integrably. The regulated principal-value prescription therefore implements a stable numerical representation of a removable singularity rather than introducing new physics.

S5.2 Order-of-magnitude scaling estimates

Two scaling statements are useful for interpreting the regime probed by SPARC.

Near-zone screening. For galactic radii $R \lesssim O(10\text{–}50)$ kpc, the near-zone condition $\kappa R \ll 1$ is satisfied for cosmologically calibrated screening scales. In this regime the Yukawa attenuation satisfies $e^{-\kappa r} \simeq 1$ across the disc, and the kernel geometry is well-approximated by its near-zone limit. Screening remains relevant through calibration (via κ fixed by the cosmological matching condition) rather than through exponential suppression on disc scales.

Bare current-sector magnitude and annulus gain. In the axisymmetric razor-thin reduction without susceptibility dressing, the induced contribution carries an explicit Lorentz suppression but also a non-local annulus-accumulation factor:

$$g_{\rm ind,bare}(R) \sim \mathcal{G}_{\rm ann}(R)\,\frac{v^2_{\rm circ}(R)}{c^2}\,g_{\rm bar}(R) \qquad (\kappa R \ll 1), \tag{S30}$$

where $\mathcal{G}_{\rm ann}(R)$ is fixed by the disc-level principal-value kernel geometry and the extent of coherent current accumulation across the disc. For representative disc rotation speeds $v \sim 100\text{–}250\,\mathrm{km\,s^{-1}}$, the Lorentz factor is

$$\frac{v^2}{c^2} \sim 10^{-7}\text{–}10^{-6},$$

but $\mathcal{G}_{\rm ann}(R)$ can reach 10^6–10^7 on the outer disc in extended systems, making $\mathcal{G}_{\rm ann}(v^2/c^2) = O(1)$. The bounded susceptibility $\mathcal{A}_{\rm eff}(R)$ then represents additional higher-order current correlations beyond the axisymmetrised truncation, rather than compensating the Lorentz suppression alone.

For NGC 3198, the outer-disc rotation-curve deficit $(V^2_{\rm obs} - V^2_{\rm bar})/(R\,g_{\rm bar})$ is approximately 3–5, so the required amplification $\mathcal{A}_{\rm eff}$ is of order 3–5 at the representative radii in Table S2, consistent with the class-level empirical amplification distribution in Fig. **??**.

Table S2. NGC 3198 annulus-gain reconciliation (illustrative radii). Values are computed from the unit-calibrated pull–drag operator used for Fig. **??**. The final column shows that $\mathcal{G}_{\rm ann}(v^2/c^2) = g_{\rm ind,bare}/g_{\rm bar}$ is $O(1)$ on the outer disc.

R (kpc)	$\mathcal{G}_{\rm ann}(R)$	$v^2_{\rm circ}/c^2$	$g_{\rm ind,bare}/g_{\rm bar}$
20.05	4.47×10^6	2.64×10^{-7}	1.18
32.14	4.03×10^6	2.40×10^{-7}	0.97
38.19	3.38×10^6	2.47×10^{-7}	0.84
44.08	2.94×10^6	2.47×10^{-7}	0.73

S5.3 Annulus-gain reconciliation: numerical example

The internal-consistency concern can be stated as follows: the Lorentz channel supplies an explicit factor v^2/c^2, but the disc operator is non-local and accumulates coherent contributions across the disc. This accumulation is captured by the dimensionless annulus-gain factor

$$\mathcal{G}_{\rm ann}(R) \;\equiv\; \frac{g_{\rm ind,bare}(R)}{\left[v^2_{\rm circ}(R)/c^2\right] g_{\rm bar}(R)}. \tag{S31}$$

Using the unit-calibrated operator export that underlies the NGC 3198 demonstrator (Fig. **??**), $\mathcal{G}_{\rm ann}(R)$ is ~few$\times 10^6$ on the outer disc. Consequently, $\mathcal{G}_{\rm ann}(v^2/c^2) = O(1)$, so the additional susceptibility $\mathcal{A}_{\rm eff}$ reported in Figs. **??** and **??** can remain modest (order unity to tens) without any 10^6–10^7 contradiction.

S5.4 Limitations of the quasi-static truncation

The SPARC audit is restricted to the conservative quasi-static sector and to a razor-thin axisymmetric reduction. The following limitations delimit the regime of validity and define the primary extension targets:

- **Neglected time derivatives and radiative sector.** The quasi-static truncation neglects $\partial_t \Phi$ and $\partial_t \mathbf{A}$ terms and therefore does not address wave-zone radiation or retardation in the conservative current sector. This is appropriate for stationary rotation-curve modelling but is not a complete treatment of time-dependent streaming or transient pattern evolution.
- **Axisymmetry and finite thickness.** The annulus operator is an axisymmetrised midplane projection. Finite vertical structure and explicit non-axisymmetric geometry alter the detailed kernel and introduce additional correlators. These effects are represented here only through the bounded susceptibility $\mathcal{A}_{\rm eff}(R)$ and are not computed from first-principles three-dimensional fields in the present audit.
- **Non-circular motions and pressure support.** Asymmetric drift corrections, warps, bars, and non-circular streaming can bias inferred circular speeds in some systems. These effects enter the analysis only through catalogue-level quality masks and the effective uncertainty model; they are not modelled explicitly in the forward closure.
- **Separation from time-asymmetric completion channels.** Time-asymmetric transport/drag completions (associated with one-way or secular effects in Machian formulations) are not used in the quasi-static force-balance closure for SPARC rotation curves. Where such channels are relevant (e.g. secular relaxation or one-way frequency-transfer observables), they require a separate treatment and are not part of the present rotation-curve mechanism.

These limitations do not affect the internal consistency of the reported quasi-static audit, but they do constrain the interpretation: the primary falsifiable extension is independent validation of the susceptibility closure through map-level estimators constructed from resolved surface-density and velocity-field data.

S5.5 Approximation budget and expected bias directions

The leading approximations used in the SPARC audit are the quasi-static truncation, razor-thin midplane reduction, axisymmetry, and the adoption of catalogue-level baryonic and geometric inputs. Their expected effects can be summarised at the level relevant for interpretation of residuals and inferred susceptibility.

Finite thickness. The razor-thin reduction neglects vertical stratification of the current source and therefore modifies the detailed annulus weighting near the midplane. The expected bias is typically to smooth the sharpest pull/drag contrasts relative to the strictly thin-disc kernel, so omission of thickness can bias the bare operator low in systems with vertically extended baryonic structure and thereby shift the inferred $\mathcal{A}_{\rm eff}$ upward.

Non-axisymmetry and coherent streaming. Bars, spirals, and lopsided structure generate low-m current correlations that are absent in the axisymmetrised closure. In the present framework these are represented phenomenologically through $\mathcal{A}_{\rm eff}(R)$. Galaxies with strong bars, marked asymmetries, or interaction signatures are therefore expected to populate the high-residual tail unless the proxy branch captures the dominant correlated structure.

Non-circular motions and pressure support. Asymmetric drift, warps, beam smearing, and local departures from circularity perturb the inferred $V_{\rm obs}(R)$ away from the circular-speed proxy. These effects are expected to broaden the residual distribution rather than mimic the smooth catalogue-level morphology of the fixed annulus operator, and are most likely to bias inferred amplification upward in gas-rich outer discs and disturbed systems.

Baryonic and geometric systematics. Distance, inclination, and stellar mass-to-light uncertainties enter the inferred acceleration budget coherently with radius. Under-estimation of baryonic surface density tends to elevate the inferred deficit and hence the required induced contribution; over-estimation acts in the opposite direction. These biases are one reason why matched nuisance treatment is essential in the Tier-2 comparison.

The practical implication is that the present audit should be read as a controlled near-zone test of operator viability rather than as a complete decomposition of all residual structure. The most likely outlier classes are therefore strongly barred systems, warped discs, galaxies with severe non-circular motions, and systems whose outer discs are especially sensitive to geometric deprojection.

S6 KERNEL DERIVATION SKETCH AND NUMERICAL PRESCRIPTION

S6.1 From screened Helmholtz to the midplane annulus kernel

The quasi-static Proca–GEM vector potential satisfies the screened Helmholtz equation (Eq. (**??**)) with source $\rho\mathbf{v}$. For an axisymmetric disc with circular streaming $\mathbf{v} = v_\phi(R)\,\hat{\boldsymbol{\phi}}$, the only component that contributes to the in-plane Lorentz channel is the azimuthal potential A_ϕ. In cylindrical coordinates and after midplane projection, the induced radial acceleration is obtained from the Lorentz-type term

$$\mathbf{a}_{\rm ind} \;\propto\; \frac{\mathbf{v}}{c}\times\mathbf{B}_g, \qquad \mathbf{B}_g = \nabla\times\mathbf{A}, \tag{S32}$$

so that the relevant field is the vertical component $B_{g,z}$ evaluated at $z = 0$.

Solving the Helmholtz equation for A_ϕ with a thin-disc source reduces to an integral over source rings. After azimuthal integration, the radial kernel depends only on the ring radii (R, R') and the relative streaming through the bilinear combination $\Delta\ell(R, R') \equiv Rv_\phi(R') - R'v_\phi(R)$, which enforces a removable singularity at co-rotation ($R' = R$). In the near-zone regime $\kappa R \ll 1$, the Yukawa attenuation is negligible over disc scales and the leading thin-disc operator takes the rational form

$$\mathcal{K}(R, R') \;\propto\; \frac{1}{R^2 - R'^2}, \tag{S33}$$

with a universal split into interior (pull) and exterior (drag) contributions according to the sign of the relative streaming. The resulting annulus operator is therefore geometry-fixed: given $\Sigma(R')$ and $v_\phi(R')$, the columns $P_{\rm bare}(R)$ and $D_{\rm bare}(R)$ are uniquely determined.

The overall normalisation is fixed by the two-body matching described in Appendix S5 and is written as $C_v = C_\star G_0/c^2$ with $C_\star$ fixed.

S6.2 Removable singularity and regulated PV prescription

The kernel contains a removable singularity at $R' = R$. In the annulus representation this appears as $(R^2 - R'^2)^{-1}$ multiplied by the signed-square bilinear in relative streaming, $\Delta\ell\,|\Delta\ell|$, which tends to zero as $R' \to R$ for smooth $v_\phi(R)$. The operator is therefore defined as a Cauchy principal value (PV). Numerically, the PV is implemented by the regulated prescription

$$(R^2 - R'^2)^{-1} \;\to\; (R^2 - R'^2 + \varepsilon^2 R^2)^{-1}, \tag{S34}$$

with fixed $\varepsilon = 10^{-3}$ under the adopted policy and explicit robustness checks (Fig. **??**). The regulator is not a physical scale; it is a numerical device for stable evaluation of a removable singularity.

S6.3 Discretisation on SPARC radii

For each galaxy, the operator is evaluated on the native SPARC radius grid $\{R_i\}$ using a one-dimensional quadrature over source radii R', with the surface-density and velocity fields taken from the SPARC baryonic mass model and the adopted streaming convention. The pull and drag columns are computed as

$$P_{\rm bare}(R_i) \equiv \sum_{R'_j<R_i} w_j\,\Sigma(R'_j)\,\mathcal{K}(R_i, R'_j)\,\Delta\ell(R_i, R'_j)\,|\Delta\ell(R_i, R'_j)|, \tag{S35}$$

$$D_{\rm bare}(R_i) \equiv \sum_{R'_j>R_i} w_j\,\Sigma(R'_j)\,\mathcal{K}(R_i, R'_j)\,\Delta\ell(R_i, R'_j)\,|\Delta\ell(R_i, R'_j)|, \tag{S36}$$

with quadrature weights w_j determined by the adopted integration rule. The induced acceleration is then $g_{\rm ind,bare}(R_i) = C_v[P_{\rm bare}(R_i) - D_{\rm bare}(R_i)]$ and the closure includes amplification via Eq. (**??**).

Convergence checks include: (i) variation of the PV regulator ε, (ii) refinement of the radial quadrature, and (iii) inner-radius smoothing where required for stability of finite-difference derivatives used in diagnostic plots.

S6.4 What is relegated to supplementary material

Longer algebraic steps in the azimuthal integration, higher-order $O((\kappa R)^2)$ corrections, and extended regulator and quadrature sweeps are deferred to supplementary material. The purpose of the present appendix is narrower: it records the minimal derivation pathway and numerical prescription needed to assess the soundness, uniqueness, and implementation-level stability of the operator used in the SPARC audit. By contrast, the supplementary material collects extended robustness checks and intermediate derivational details that are not required for following the main logic of the paper.

S7 REPRODUCIBLE OPERATOR EXPORTS AND METADATA

To support independent verification of the thin-disc current operator, a machine-readable export is provided for the NGC 3198 demonstrator. The export consists of (i) a tabulated operator file containing the radius grid and the pull/drag operator columns, and (ii) a compact metadata sidecar specifying the constants and unit conventions required to reconstruct the induced acceleration without reference to rotation-curve residuals.

Matched halo-baseline provenance. The Burkert baseline reported in the matched Tier 1 and Tier 2 comparisons was generated by a protocol-identical substitution of a standard two-parameter Burkert halo for the halo component in the released NFW workflow, rather than by a separately archived native Burkert branch. In particular, the Burkert runs follow the same retained SPARC sample, the same point mask $V_{\rm obs}/\sigma_V \geq 3$, the same effective-uncertainty prescription $\sigma^2_{\rm eff} = \sigma^2_V + (0.2V_{\rm obs})^2$ with $0.2 = 0.2$, and the same information-criterion accounting as the corresponding NFW comparisons. For Tier 2, the Burkert branch follows the same MAP nuisance-treatment model as the released Tier-2 script, including the same parameter structure (s_D, i, Υ), the same Gaussian/log-normal prior class, and the same ROTMOD-header/omnibus distance and inclination heuristics recorded in the Tier-2 manifest; only the halo profile itself is changed. The resulting Burkert totals should therefore be interpreted as matched baseline results under the stated protocol, not as outputs from a distinct optimisation or a different sample definition.

S7.1 Operator export contents

The operator export is provided as a comma-separated values (CSV) table evaluated on the native SPARC radius grid. The file contains, at minimum, the following columns:

- R (kpc): galactocentric radius;
- $P_{\rm bare}(R)$ and $D_{\rm bare}(R)$: inner and outer annulus operator columns evaluated under the regulated PV prescription of Appendix S6.2;
- $\Delta(R) \equiv P_{\rm bare}(R) - D_{\rm bare}(R)$: net operator column used in the induced term;
- $g_{\rm ind,bare}(R)$ (optional): induced acceleration column, included only if the prefactor has been applied.

The export is intended to permit direct reconstruction of the induced term used in the closure,

$$g_{\rm ind}(R) = \mathcal{A}_{\rm eff}(R)\, C_v\, [P_{\rm bare}(R) - D_{\rm bare}(R)], \tag{S37}$$

with C_v fixed by two-body matching and with amplification treated according to the adopted policy.

In the public archive, these products are organised as `operator_export_ngc3198.csv`, `operator_export_ngc3198_metadata.yaml`, `tier1_matched_totals.csv`, `tier2_matched_totals.csv`, `pilot_resolved_map_summary.csv`, and the associated plotting and configuration scripts; exact filenames should match the release tag named in the Data Availability statement.

S8 MAP-LEVEL VALIDATION PROTOCOL FOR THE AMPLIFICATION SUSCEPTIBILITY

This appendix summarises a survey-ready protocol for estimating $\widehat{\mathcal{A}}_{\rm eff}(R)$ directly from resolved surface-density and velocity-field data, thereby defining an external test of the susceptibility closure that does not rely on rotation-curve residuals.

A minimal implementation requires matched resolved surface-density maps (H I, and optionally CO), a line-of-sight velocity field with an associated mask, and disc geometry (inclination, position angle, systemic velocity) consistent with the rotation-curve construction; public examples include THINGS and PHANGS products (Walter et al. 2008; de Blok et al. 2008; Trachternach et al. 2008; Leroy et al. 2021). After deprojection, one forms the surface mass current $j_\phi(R,\phi) \equiv \Sigma(R,\phi)\, v_\phi(R,\phi)$, its azimuthal mean $\bar{j}_\phi(R)$, and low-order Fourier modes

$$j_{\phi,m}(R) \equiv \frac{1}{2\pi}\int_0^{2\pi} j_\phi(R,\phi)\, e^{-im\phi}\, d\phi, \qquad m \geq 0, \tag{S38}$$

so that $j_{\phi,0}(R) = \bar{j}_\phi(R)$. Using the same thin-disc geometric kernel $\mathcal{K}(R,R')$ and regulated principal-value prescription as in the disc-integrated operator, an operational estimator of the amplification is

$$\widehat{\mathcal{A}}_{\rm eff}(R) \equiv \frac{\mathcal{N}(R)}{\mathcal{D}(R)}, \tag{S39}$$

$$\mathcal{N}(R) = \int dR'\, \mathcal{K}(R,R') \left[\bar{j}_\phi(R)\bar{j}_\phi(R') + \sum_{m=1}^{m_{\max}} j_{\phi,m}(R)\, j^*_{\phi,m}(R') \right], \tag{S40}$$

$$\mathcal{D}(R) = \int dR'\, \mathcal{K}(R,R')\, \bar{j}_\phi(R)\bar{j}_\phi(R'). \tag{S41}$$

In practice, $m_{\max} = 2$ or 4 is usually sufficient when bars and grand-design spirals dominate. By construction, $\widehat{\mathcal{A}}_{\rm eff}(R) = 1$ for a strictly axisymmetric laminar disc.

NGC 3198 is a natural demonstrator for the external susceptibility programme because it is both well resolved and archetypal at the level of the rotation-curve phenomenology itself. Before turning to the map-level estimator, Fig. S1 records the corresponding RAR-surrogate closure for this system, illustrating in compact form the observed lift from the baryonic Newtonian prediction into the flat outer-disc regime. This figure is supplementary to the main-text bare-operator and pull–drag diagnostics: it is included here only to motivate why NGC 3198 remains the reference case for the resolved-map proof-of-concept.

For NGC 3198, the protocol can be applied using publicly available THINGS products together with the adopted SPARC geometry. In the present demonstrator, the deprojection uses the Walter et al. (2008) centre, distance, inclination, and position angle; valid-pixel support is derived from the associated THINGS cube; and the velocity zero-point is taken from the cube header because no standalone downloadable systemic-velocity product was identified in the working data bundle. The resulting $\widehat{\mathcal{A}}_{\rm eff}(R)$ profile is therefore an external prediction target for the susceptibility closure. In the present draft, the map-derived estimator is shown on its own, while direct overlay against the proxy-policy profile of Eq. (**??**) is deferred until the manuscript's final shared-fit coefficients are fixed numerically.

For completeness, Fig. S3 provides a compact visual summary of the representative-galaxy table in the main text. The figure is not intended to replace the full-sample structural trend of Fig. **??**; it simply shows how the numeric outer-disc amplification proxy $\widetilde{\mathcal{A}}^{\rm out}_{\rm eff}$ distributes across the closure classes for the named systems listed in Table **??**.

S8.1 Admissible closure family and operational meaning of "bounded"

In the present manuscript, the term "bounded" is used in a restricted operational sense rather than as a claim of a fully derived universal saturation theorem. The amplification factor $\mathcal{A}_{\rm eff}(R)$ is not treated as an arbitrary per-galaxy function of radius, but as a member of a prescribed closure family whose radial dependence is supplied by externally defined structural fields or templates and whose amplitude is controlled only by explicitly stated shared coefficients. The relevant restriction is therefore the admissible range of the declared closure under the stated coefficient policy and proxy inputs, not an unconstrained halo-like profile.

This point matters for interpretation of the SPARC fits. The closure used here does not permit galaxy-by-galaxy radial inflation of $\mathcal{A}_{\rm eff}(R)$ in response to residual structure. Any inferred profile must remain within the chosen proxy family, and its physical plausibility is subject to external appraisal through resolved current correlators and independent streaming diagnostics. A first-principles universal ceiling from nonlinear mode saturation is not claimed to have been derived in the present paper.

S8.2 Scope of the present resolved-map appraisal

The present manuscript does not claim a completed catalogue-level validation of the susceptibility closure. The resolved-map construction is included to define a falsifiable external target and to demonstrate, using NGC 3198 as a proof-of-concept, how the required estimator can be assembled from public survey products and the same regulated thin-disc kernel used in the SPARC audit. A broader programme applying this estimator to a suitably selected set of nearby spirals with homogeneous kinematic maps would be the appropriate next step for testing whether the inferred proxy-branch amplitudes are reproduced by directly measured current structure.

S8.3 Metadata and reconstruction

A compact metadata sidecar (JSON or YAML) accompanies the operator CSV and records the physical constants and fixed coefficients (G_0, c, $C_\star$, C_ν), the screening parameter κ and declared regime $\kappa R \ll 1$, the numerical prescription (PV regulator ε, quadrature rule, and any inner-radius smoothing rule), the unit conventions for surface density, velocity, and radius, the prefactor status (whether C_ν has already been applied), and the provenance of the export. These items are sufficient to reconstruct the induced contribution in absolute units from the exported columns without anchor normalisation or reference to SPARC residuals.

Given the operator CSV and metadata sidecar, reconstruction proceeds by forming $\Delta(R) = P_{\rm bare}(R) - D_{\rm bare}(R)$, applying C_ν if needed to obtain $g_{\rm ind,bare}(R)$, multiplying by $\mathcal{A}_{\rm eff}(R)$ if an amplification branch is active, and then constructing $V^2_{\rm model}(R) = V^2_{\rm bar}(R) + R\, g_{\rm ind}(R)$ under the stated $\Upsilon_\star$ convention. The operator morphology diagnostic (Fig. **??**) provides an internal consistency check on the exported columns, and the regulator audit (Fig. **??**) verifies numerical stability of the PV prescription. A successful outcome would be agreement in radial scale and gross morphology within the observational and deprojection uncertainties; a systematic mismatch would count against the susceptibility interpretation even if the rotation-curve fit remained acceptable.

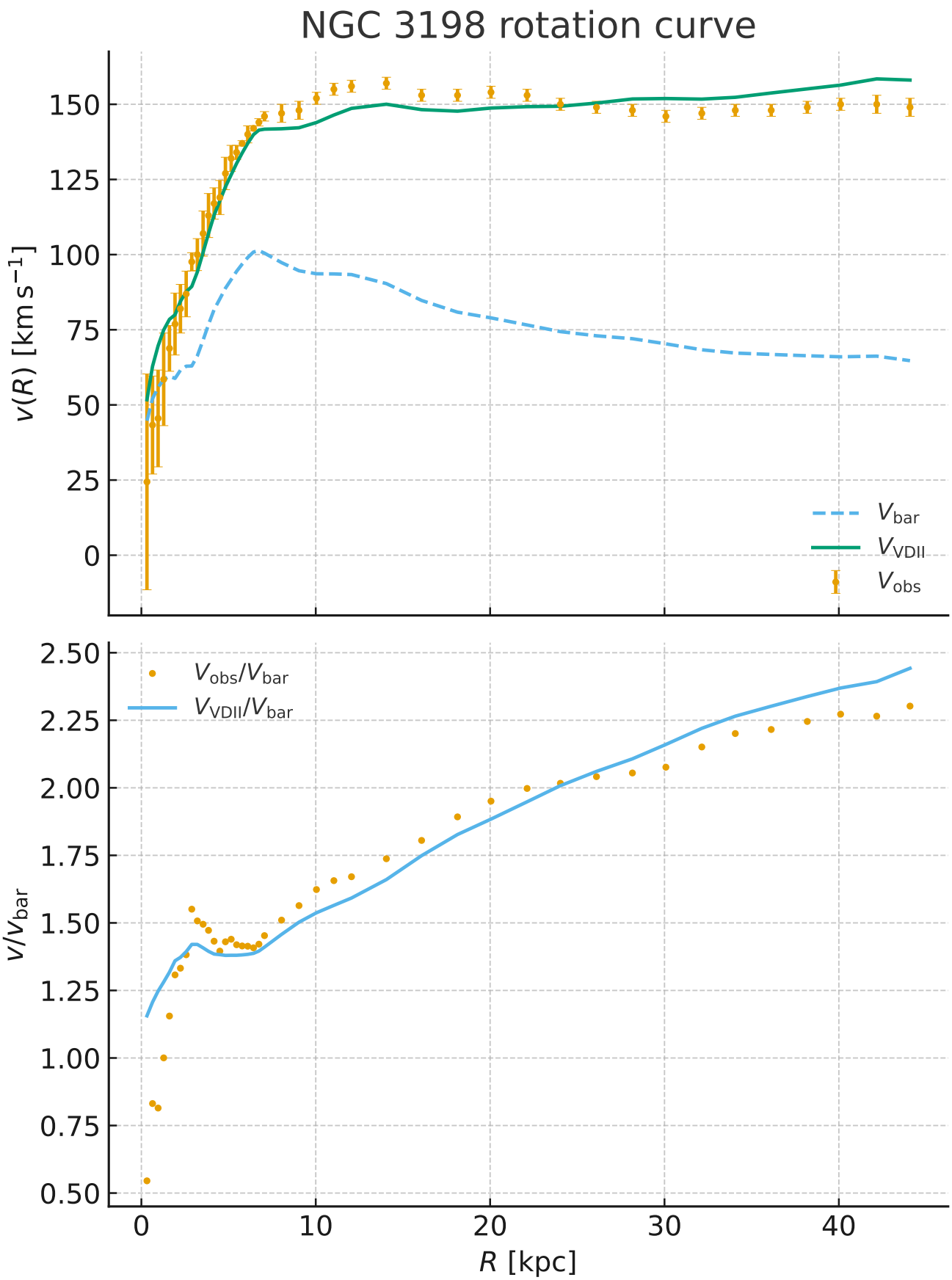

Figure S1. NGC 3198 as an archetypal flat-curve demonstrator (RAR-surrogate closure). Upper panel: observed rotation curve with SPARC uncertainties, Newtonian baryonic prediction $V_{\rm bar}(R)$ and the surrogate VDII closure prediction. Lower panel: the enhancement ratio $V(R)/V_{\rm bar}(R)$ illustrating the lift from the baryonic curve into the observed flat outer regime. This supplementary figure is included to motivate the choice of NGC 3198 as the reference system for the resolved-map proof-of-concept; the main text uses the unit-calibrated bare-operator closure and pull–drag shape diagnostics instead.

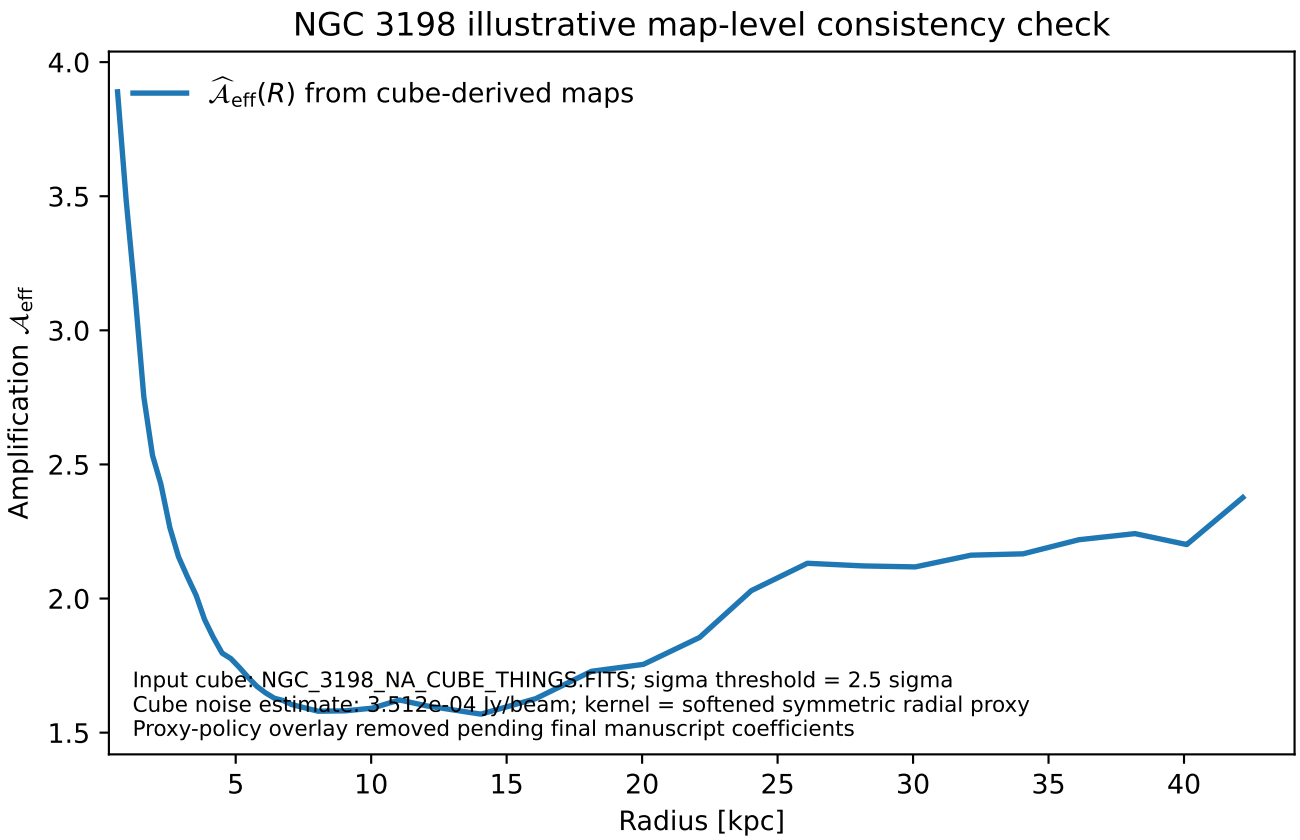

Figure S2. NGC 3198 illustrative map-level consistency check of the susceptibility closure. Cube-derived H I moment information for NGC 3198 is used to construct a direct estimator $\widehat{\mathcal{A}}_{\rm eff}(R)$ on annuli defined using the adopted THINGS geometry, with valid-pixel support derived from the associated data cube and a header-derived velocity zero-point. The figure is intended as an external map-level diagnostic, independent of the rotation-curve residual fit. In the present draft only the map-derived estimator is shown; a direct overlay of the proxy-policy branch is deferred until the manuscript's final shared-fit coefficients are fixed numerically.

S9 TIER-1 MATCHED-REFIT BASELINE (FIXED BARYONS, FIXED GEOMETRY)

S10 TIER-1 MATCHED-REFIT BASELINE (FIXED BARYONS, FIXED GEOMETRY)

Under the common Tier-1 protocol, Table S3 gives the matched ranking on the same 175-galaxy SPARC subset ($N = 3317$). Relative to the baryonic-only reference, the one-parameter VDII surrogate substantially improves the fit and outperforms the matched MOND surrogate. The fixed-baryon NFW and Burkert halo baselines achieve much lower raw χ^2, with Burkert also giving the lowest AIC, but both do so at the cost of 350 fitted halo parameters. Under the stronger complexity penalty of BIC, the VDII surrogate remains preferred. The Tier-1 comparison therefore highlights a likelihood–parsimony trade-off under a common mask and likelihood rather than a stand-alone proof of mechanism.

Over $N = 3317$ retained points across 175 galaxies, the baryonic-only baseline gives $\chi^2 = 9212.19$ with $\chi^2_{\rm red} = 2.777$, providing the expected poor reference fit in the absence of any non-baryonic or effective-amplification support. The VDII RAR-surrogate reduces this to $\chi^2 = 4869.81$ with $\chi^2_{\rm red} = 1.469$, AIC = 4871.81, and BIC = 4877.92. The matched MOND surrogate gives $\chi^2 = 5112.72$ with $\chi^2_{\rm red} = 1.542$, AIC = 5114.72, and BIC = 5120.83. The fixed-baryon NFW baseline attains the lowest raw χ^2, namely $\chi^2 = 2910.75$ with $\chi^2_{\rm red} = 0.981$, but only by using 350 fitted halo parameters, which raises the complexity-penalised criteria to AIC = 3610.75 and BIC = 5748.13. Thus, under identical masking and likelihood treatment, the VDII surrogate improves dramatically over the baryonic-only baseline, is decisively preferred over the matched MOND surrogate, and remains strongly favoured over the heavily parameterised NFW baseline by BIC even though NFW wins in raw likelihood.

This comparison demonstrates matched statistical efficiency under the adopted protocol, and does not represent a stand-alone proof of the underlying mechanism.

This appendix reports the Tier-1 matched-refit baseline used to contextualise the main Tier-2 comparison. Tier 1 enforces a deliberately minimal and transparent refit policy: a common point mask ($V_{\rm obs}/\sigma_V \geq 3$), a common effective uncertainty prescription ($\sigma^2_{\rm eff} = \sigma^2_V + (0.2V_{\rm obs})^2$ with 0.2 = 0.2), fixed baryonic mass models under the SPARC convention, and fixed geometry (no refitting of distance or inclination). Under this policy, the VDII RAR-surrogate and MOND comparators each introduce one shared global scale, while the NFW baseline introduces two halo parameters per galaxy. The resulting catalogue-level χ^2, AIC and BIC values are summarised in the Tier-1 table, with Figure S4

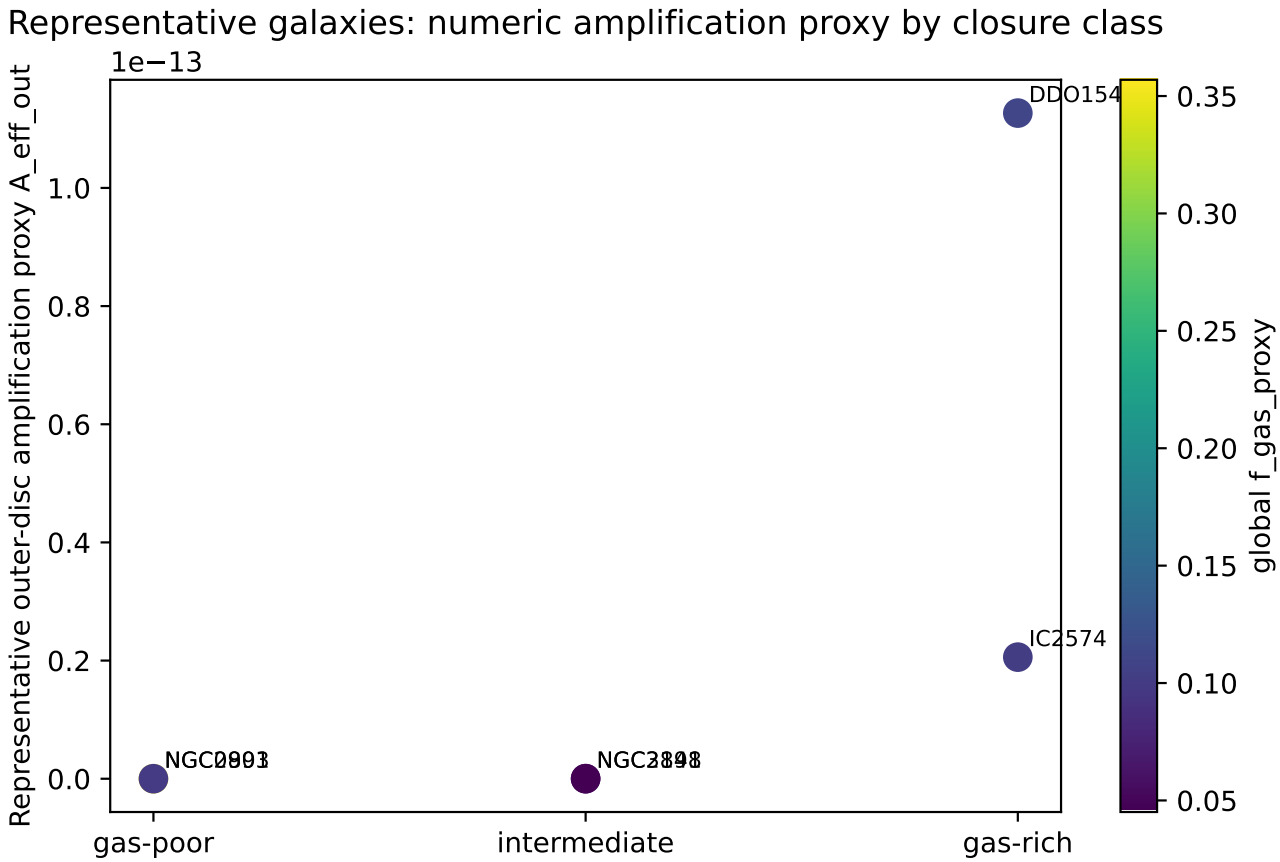

Figure S3. Representative galaxies with numeric outer-disc amplification proxies. The representative per-galaxy amplification proxy $\widetilde{\mathcal{A}}_{\rm eff}^{\rm out}$ from Table **??** is shown by closure class, with point colour indicating the global gas-fraction proxy $f_{\rm gas,proxy} = M_{\rm HI}/(M_{\rm HI} + L_{3.6})$. This figure is intended as a compact visual companion to the table rather than a replacement for the full catalogue-level structural trend of Fig. **??**.

Table S3. Matched Tier-1 comparison on the common SPARC subset. All models are evaluated on the same 175-galaxy subset with $N = 3317$ retained points under the common Tier-1 prescription. Here k is the fitted parameter count under the adopted accounting convention.

Model	k	χ^2	$\chi^2_{\rm red}$	AIC	BIC
Baryonic	0	9212.19	2.777	9212.19	9212.19
VDII	1	4869.81	1.469	4871.81	4877.92
MOND	1	5112.72	1.542	5114.72	5120.83
NFW	350	2910.75	0.981	3610.75	5748.13
Burkert	350	2552.37	0.860	3252.37	5389.76

providing the corresponding information-criterion comparison; the per-galaxy $\chi^2_{\rm red}$ distribution is retained as a supplementary diagnostic. Tier 1 is not intended to define the preferred inference protocol; rather, it provides a fixed-geometry reference point for diagnosing how much of the comparative performance is attributable to nuisance freedom versus the structural form of each dynamical closure.

Glossary of selected theoretical and methodological terminology

Proca–GEM Screened gravitoelectromagnetic current sector formulated as a massive vector (Proca) field with Helmholtz-type potentials in the quasi-static limit.

Yukawa screening Exponential attenuation of the screened Helmholtz potentials over the length scale κ^{-1}; negligible on SPARC disc scales in the near-zone regime $\kappa R \ll 1$, but retained as part of the parent theory and its calibration.

Pull–drag operator The razor-thin annulus reduction in which interior and exterior source annuli contribute with opposite tangential sense to the induced radial term.

Servomechanism Broader velocity-dependent inertial-induction picture in which pull–drag interactions tend toward dynamical reorganisation over long timescales.

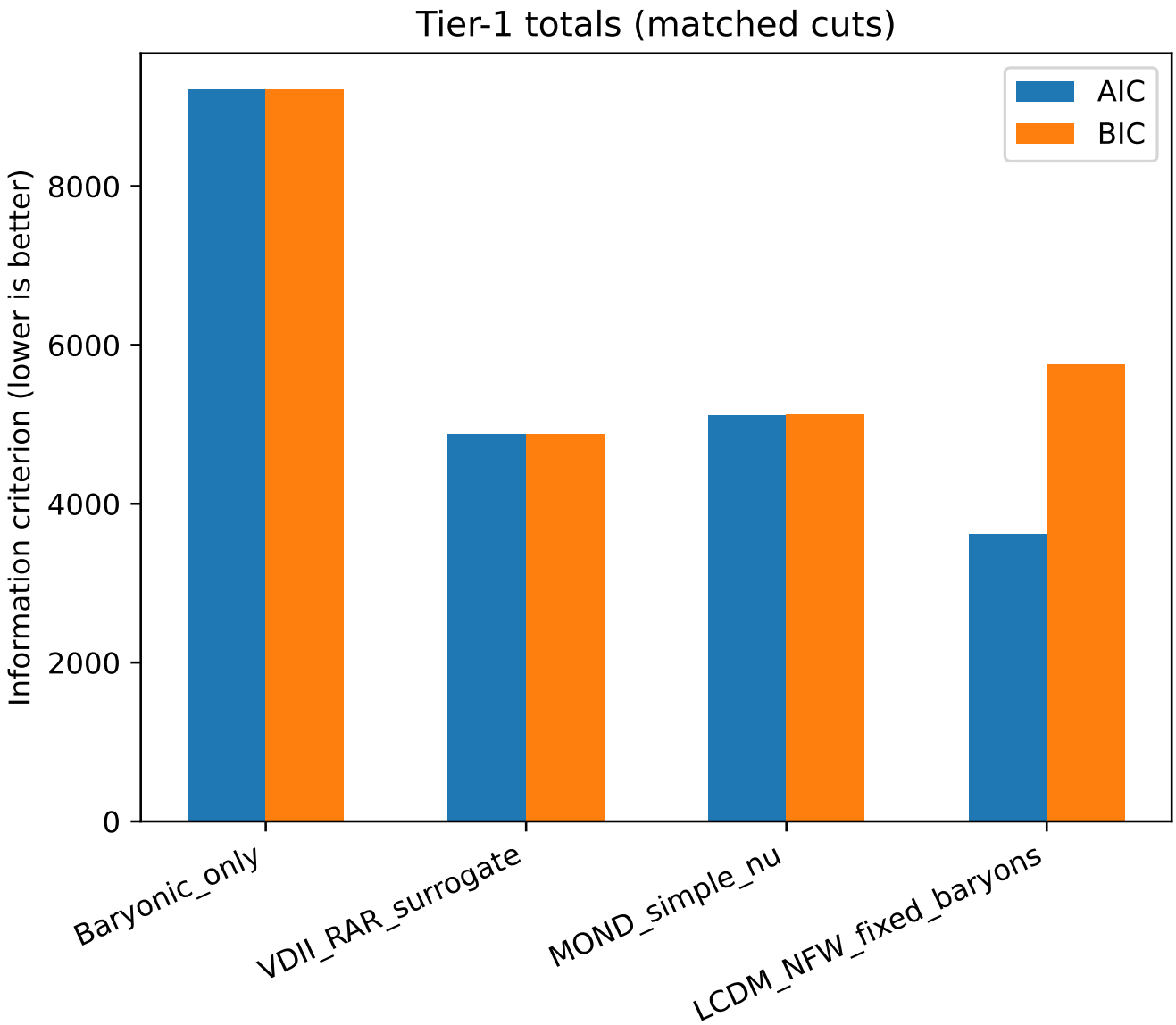

Figure S4. Tier-1 information-criterion comparison under fixed-geometry refits (SPARC). The figure summarises AIC and BIC values from the Tier-1 protocol, evaluated under the common point mask $V_{\rm obs}/\sigma_V \geq 3$ and the common effective uncertainty prescription $\sigma_{\rm eff}^2 = \sigma_V^2 + (0.2V_{\rm obs})^2$ with 0.2 = 0.2. The matched comparison includes the baryonic-only baseline, the VDII RAR surrogate, the matched MOND simple-ν surrogate, and the fixed-baryon NFW and Burkert halo baselines. Lower values indicate the preferred balance between likelihood and parameter count.

Sciama–Ghosh inertial induction Velocity-dependent inertial-induction framework used here as the two-body antecedent of the disc-level pull–drag operator and its normalisation.

Principal-value regulator (ε) Numerical regularisation used to evaluate the removable co-rotation singularity in the annulus operator at $R' = R$.

Intrinsic astrophysical scatter (0.2) Additional fractional term in the effective velocity uncertainty, used to absorb unresolved kinematic structure and other catalogue-level systematics.

Susceptibility / amplification ($\mathcal{A}_{\rm eff}$) Bounded closure factor multiplying the bare operator to represent omitted higher-order current correlations; in this paper its radial dependence is supplied by externally defined proxy fields or templates rather than by per-galaxy fitted radial functions.

Two-body matching Procedure fixing the kernel normalisation $C_\nu = C_\star G_0/c^2$ by matching the annulus operator to the corresponding two-body inertial-induction interaction.

Mode A Unit-calibrated bare-operator evaluation used for kernel diagnostics, sign checks, and regulator robustness tests.

Mode B Closure-mode SPARC analysis in which the induced contribution is compared with the catalogue-level rotation-curve phenomenology and class-normalised amplification profiles.

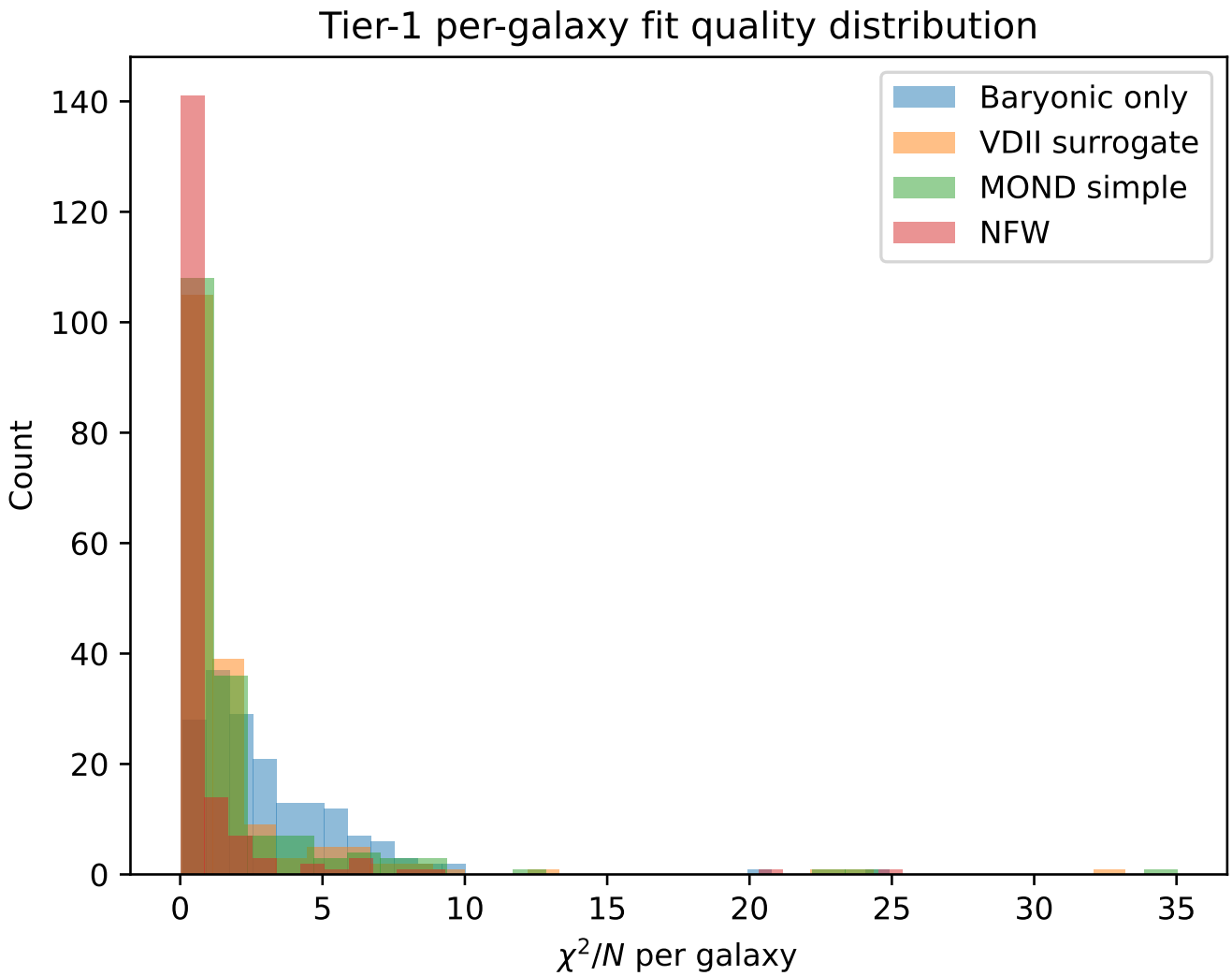

Figure S5. Distribution of per-galaxy reduced χ^2 under Tier-1 refits. Histograms show per-galaxy reduced χ^2 values under the same Tier-1 protocol and point mask as Table S3. The matched comparison includes the baryonic-only baseline, the VDII RAR surrogate, the matched MOND simple-ν surrogate, and the fixed-baryon NFW and Burkert halo baselines. This diagnostic complements the global information criteria by showing how fit quality is distributed across the sample rather than only in the total likelihood.

RAR-surrogate closure One-parameter analytic closure used as a catalogue-level proxy for the low-acceleration transition, with a single global scale $g_\dagger$.

Resolved-map estimator External estimator $\widehat{\mathcal{A}}_{\rm eff}(R)$ constructed from resolved surface-density and velocity-field data to test the susceptibility interpretation independently of rotation-curve fitting.

S11 REPRODUCIBILITY TABLES AND UNCERTAINTY BUDGET

This appendix collects the compact methodological tables requested for reproducibility of the matched SPARC audit. Machine-readable versions of the full sample-summary and per-galaxy matched-refit tables should accompany the public archive described in the Data Availability statement.

S11.1 Compact systematic-uncertainty budget

[t]

Table S4. Compact systematic-uncertainty budget for the SPARC audit and resolved-map pilot. Entries state whether a source enters a given analysis branch and how it is treated under the adopted policy.

Source	Tier 1	Tier 2	Map pilot	Treatment under the adopted policy
Velocity measurement error σ_V	included	included	not used in this form	Enters the rotation-curve likelihood through $\sigma_{\rm eff}^2 = \sigma_V^2 + (0.2V_{\rm obs})^2$.
Intrinsic scatter $\sigma_{\rm int}$	fixed	fixed	not used in this form	Held fixed at 0.2 = 0.2 in the rotation-curve likelihood; not an optimised parameter in the present draft.
Distance	fixed catalogue value	optimised with prior	propagated	Held fixed in Tier 1; varied via MAP nuisance optimisation in Tier 2; propagated through deprojection resampling in the map pilot.
Inclination	fixed catalogue value	optimised with prior	propagated	Held fixed in Tier 1; varied via MAP nuisance optimisation in Tier 2; propagated through deprojection resampling in the map pilot.
Stellar M/L	fixed SPARC convention	optimised with SPS prior	not directly fitted	Held fixed in Tier 1; varied in Tier 2 only within the stated SPS prior; enters map interpretation indirectly through the adopted baryonic decomposition.
Bulge normalisation	fixed SPARC convention	optimised with prior where relevant	not directly fitted	Fixed in Tier 1; varied only for galaxies with a bulge component and only under the stated prior in Tier 2.
Beam smearing / non-circular motions	not explicitly propagated	not explicitly propagated	propagated / diagnosed	Absorbed only indirectly into the residual budget in Tier 1 and Tier 2; treated explicitly through masking, beam support, and deprojection/error propagation in the map pilot.
Map deprojection uncertainty	not applicable	not applicable	propagated	Monte Carlo resampling of centre, position angle, inclination, systemic velocity, masking, and support thresholds in the resolved-map estimator.

S11.2 Tier-2 nuisance priors

[t]

Table S5. Tier-2 nuisance priors used in the matched-refit audit. Exact functional forms and bounds match the released Tier-2 optimisation script.

Parameter	Prior family	Centre	Width	Bounds / remarks
Distance scale s_D	Gaussian in s_D	1	$D_\sigma \equiv e_D/D$	Truncated to $s_D \in [\max(0.5,\ 1-3D_\sigma),\ \min(1.5,\ 1+3D_\sigma)]$ on the optimisation grid; applied per galaxy using the inferred catalogue/header distance and distance uncertainty.
Inclination i	Gaussian in i (degrees)	$i_{\rm nom}$	$i_\sigma \equiv \max(e_i, 10^{-3})$	Applied directly in i (degrees), not in $\cos i$ or $\sin i$; truncated to $i \in [\max(5^\circ,\ i_{\rm nom}-3i_\sigma),\ \min(89^\circ,\ i_{\rm nom}+3i_\sigma)]$. The Tier-2 heuristic sets the nominal inclination and its uncertainty as follows: if Inc ≤ 5, interpret Inc as σ_i and $e_{\rm Inc}$ as $i_{\rm nom}$; otherwise take $i_{\rm nom}$ = Inc and $i_\sigma = e_{\rm Inc}/10$, with fallback clipping where needed.
Disc $\Upsilon_\star$	Log-normal in $\Upsilon_\star$ (Gaussian in $\ln\Upsilon_\star$)	$\ln\Upsilon_\star = 0$ ($\Upsilon_\star = 1$)	$\sigma_{\ln\Upsilon} = 0.3$	Applied in logarithmic space. The optimisation grid uses $\Upsilon_\star \in [\exp(-3\sigma_{\ln\Upsilon}),\ \exp(+3\sigma_{\ln\Upsilon})] = [\exp(-0.9),\ \exp(0.9)] \approx [0.407,\ 2.460]$.
Bulge $\Upsilon_{\rm bul}$ or bulge normalisation	Not separately implemented	—	—	No distinct bulge nuisance parameter is introduced in the released Tier-2 code. Instead, the same log-normal $\Upsilon_\star$ scale factor is applied jointly to the disc and bulge contributions through $V_{\rm disk} \mapsto \sqrt{\Upsilon_\star}\,V_{\rm disk}$ and $V_{\rm bul} \mapsto \sqrt{\Upsilon_\star}\,V_{\rm bul}$. Any future separate bulge prior should then be applied uniformly across VDII, MOND, NFW, and Burkert baselines.

S11.3 Sample summary for the retained SPARC subset

The full retained-sample table is given below in long-table form so that it can break cleanly across pages while preserving column headings.

Table S6: **Sample summary for the retained SPARC subset used in the matched comparisons.** Columns list the galaxy identifier, the adopted SPARC quality flag Q, the catalogue inclination, the number of retained points under the common mask, the number of excluded points, and membership in the Tier 1, Tier 2, and resolved-map-pilot subsets. The full table should be released in machine-readable form.

Galaxy	Q	i [°]	$N_{\rm retained}$	$N_{\rm removed}$	Tier 1	Tier 2	Map pilot
CamB	1	62.0	8	1	✓	✓	–
D512-2	0	61.0	4	0	✓	✓	–
D564-8	0	82.0	6	0	✓	✓	–
D631-7	0	82.0	16	0	✓	✓	–
DDO064	3	41.0	13	1	✓	✓	–
DDO154	5	2.0	12	0	✓	✓	–
DDO161	1	51.0	30	1	✓	✓	–
DDO168	4	12.0	10	0	✓	✓	–
DDO170	9	21.0	8	0	✓	✓	–
ESO079-G014	8	71.0	14	1	✓	✓	–
ESO116-G012	10	1.0	15	0	✓	✓	–
ESO444-G084	3	82.0	7	0	✓	✓	–
ESO563-G021	6	1.0	29	1	✓	✓	–
F561-1	1	1.0	5	1	✓	✓	–
F563-1	3	1.0	17	0	✓	✓	–
F563-V1	7	1.0	5	1	✓	✓	–
F563-V2	1	1.0	7	3	✓	✓	–
F565-V2	7	1.0	6	1	✓	✓	–
F567-2	5	1.0	5	0	✓	✓	–
F568-1	6	1.0	11	1	✓	✓	–
F568-3	6	41.0	17	1	✓	✓	–
F568-V1	4	61.0	13	2	✓	✓	–
F571-8	4	1.0	12	1	✓	✓	–
F571-V1	0	1.0	7	0	✓	✓	–
F574-1	6	81.0	14	0	✓	✓	–
F574-2	3	11.0	3	2	✓	✓	–
F579-V1	1	51.0	14	0	✓	✓	–
F583-1	6	51.0	20	5	✓	✓	–
F583-4	8	1.0	11	1	✓	✓	–
IC2574	1	2.0	34	0	✓	✓	–
IC4202	2	1.0	32	0	✓	✓	–
KK98-251	3	41.0	14	1	✓	✓	–
NGC0024	7	62.0	28	1	✓	✓	–
NGC0055	0	12.0	21	0	✓	✓	–
NGC0100	6	51.0	20	1	✓	✓	–
NGC0247	3	92.0	26	0	✓	✓	–
NGC0289	10	1.0	28	0	✓	✓	–
NGC0300	9	2.0	25	0	✓	✓	–
NGC0801	5	71.0	13	0	✓	✓	–
NGC0891	8	2.0	18	0	✓	✓	–
NGC1003	3	21.0	36	0	✓	✓	–
NGC1090	0	51.0	23	1	✓	✓	–
NGC1705	2	92.0	14	0	✓	✓	–
NGC2366	7	62.0	25	1	✓	✓	–
NGC2403	5	62.0	73	0	✓	✓	–
NGC2683	4	92.0	11	0	✓	✓	–
NGC2841	5	3.0	50	0	✓	✓	–
NGC2903	4	81.0	34	0	✓	✓	–
NGC2915	7	2.0	27	3	✓	✓	–
NGC2955	0	1.0	24	0	✓	✓	–
NGC2976	3	82.0	25	2	✓	✓	–

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Table S6: **Sample summary for the retained SPARC subset used in the matched comparisons (continued).**

Galaxy	Q	i [°]	$N_{\rm retained}$	$N_{\rm removed}$	Tier 1	Tier 2	Map pilot
NGC2998	4	1.0	13	0	✓	✓	–
NGC3109	6	72.0	25	0	✓	✓	–
NGC3198	6	3.0	40	3	✓	✓	✓
NGC3521	9	1.0	41	0	✓	✓	–
NGC3726	2	4.0	12	0	✓	✓	–
NGC3741	4	72.0	21	0	✓	✓	–
NGC3769	2	4.0	12	0	✓	✓	–
NGC3877	1	4.0	13	0	✓	✓	–
NGC3893	1	4.0	10	0	✓	✓	–
NGC3917	4	4.0	17	0	✓	✓	–
NGC3949	4	4.0	7	0	✓	✓	–
NGC3953	7	4.0	8	0	✓	✓	–
NGC3972	0	4.0	10	0	✓	✓	–
NGC3992	3	5.0	9	0	✓	✓	–
NGC4010	8	4.0	12	0	✓	✓	–
NGC4013	1	4.0	36	0	✓	✓	–
NGC4051	5	4.0	7	0	✓	✓	–
NGC4068	3	22.0	5	1	✓	✓	–
NGC4085	0	4.0	7	0	✓	✓	–
NGC4088	2	4.0	12	0	✓	✓	–
NGC4100	8	4.0	24	0	✓	✓	–
NGC4138	8	4.0	7	0	✓	✓	–
NGC4157	4	4.0	17	0	✓	✓	–
NGC4183	6	4.0	23	0	✓	✓	–
NGC4214	6	42.0	14	0	✓	✓	–
NGC4217	7	4.0	19	0	✓	✓	–
NGC4389	7	4.0	6	0	✓	✓	–
NGC4559	1	1.0	32	0	✓	✓	–
NGC5005	0	5.0	18	0	✓	✓	–
NGC5033	10	1.0	22	0	✓	✓	–
NGC5055	5	2.0	28	0	✓	✓	–
NGC5371	0	21.0	19	0	✓	✓	–
NGC5585	1	21.0	24	0	✓	✓	–
NGC5907	0	2.0	19	0	✓	✓	–
NGC5985	10	1.0	33	0	✓	✓	–
NGC6015	1	1.0	44	0	✓	✓	–
NGC6195	1	1.0	23	0	✓	✓	–
NGC6503	4	12.0	31	0	✓	✓	–
NGC6674	0	1.0	14	1	✓	✓	–
NGC6789	1	82.0	4	0	✓	✓	–
NGC6946	1	62.0	58	0	✓	✓	–
NGC7331	7	3.0	36	0	✓	✓	–
NGC7793	7	82.0	43	3	✓	✓	–
NGC7814	2	62.0	18	0	✓	✓	–
PGC51017	4	2.0	6	0	✓	✓	–
UGC00128	1	1.0	21	1	✓	✓	–
UGC00191	0	1.0	9	0	✓	✓	–
UGC00634	7	1.0	4	0	✓	✓	–
UGC00731	2	51.0	12	0	✓	✓	–
UGC00891	6	1.0	5	0	✓	✓	–
UGC01230	6	1.0	10	1	✓	✓	–
UGC01281	5	42.0	20	5	✓	✓	–
UGC02023	6	1.0	4	1	✓	✓	–
UGC02259	7	1.0	8	0	✓	✓	–
UGC02455	7	81.0	8	0	✓	✓	–
UGC02487	0	1.0	17	0	✓	✓	–
UGC02885	4	61.0	19	0	✓	✓	–

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Table S6: **Sample summary for the retained SPARC subset used in the matched comparisons (continued).**

Galaxy	Q	i [°]	N_{retained}	N_{removed}	Tier 1	Tier 2	Map pilot
UGC02916	4	1.0	43	0	✓	✓	–
UGC02953	4	51.0	115	0	✓	✓	–
UGC03205	9	1.0	48	0	✓	✓	–
UGC03546	8	1.0	30	0	✓	✓	–
UGC03580	10	1.0	45	2	✓	✓	–
UGC04278	9	51.0	23	2	✓	✓	–
UGC04305	7	72.0	21	1	✓	✓	–
UGC04325	7	81.0	8	0	✓	✓	–
UGC04483	1	12.0	7	1	✓	✓	–
UGC04499	9	51.0	9	0	✓	✓	–
UGC05005	2	1.0	8	3	✓	✓	–
UGC05253	7	21.0	73	0	✓	✓	–
UGC05414	7	21.0	6	0	✓	✓	–
UGC05716	1	1.0	12	0	✓	✓	–
UGC05721	7	51.0	23	0	✓	✓	–
UGC05750	7	1.0	8	3	✓	✓	–
UGC05764	4	41.0	10	0	✓	✓	–
UGC05829	8	91.0	10	1	✓	✓	–
UGC05918	6	1.0	8	0	✓	✓	–
UGC05986	6	91.0	15	0	✓	✓	–
UGC05999	5	1.0	5	0	✓	✓	–
UGC06399	9	4.0	9	0	✓	✓	–
UGC06446	0	1.0	17	0	✓	✓	–
UGC06614	1	71.0	13	0	✓	✓	–
UGC06628	0	31.0	7	0	✓	✓	–
UGC06667	8	4.0	9	0	✓	✓	–
UGC06786	0	21.0	45	0	✓	✓	–
UGC06787	6	21.0	71	0	✓	✓	–
UGC06818	7	4.0	8	0	✓	✓	–
UGC06917	3	4.0	11	0	✓	✓	–
UGC06923	7	4.0	6	0	✓	✓	–
UGC06930	7	4.0	10	0	✓	✓	–
UGC06973	2	4.0	9	0	✓	✓	–
UGC06983	6	4.0	17	0	✓	✓	–
UGC07089	9	4.0	12	0	✓	✓	–
UGC07125	3	1.0	13	0	✓	✓	–
UGC07151	6	42.0	11	0	✓	✓	–
UGC07232	2	72.0	4	0	✓	✓	–
UGC07261	0	31.0	7	0	✓	✓	–
UGC07323	7	1.0	10	0	✓	✓	–
UGC07399	8	31.0	10	0	✓	✓	–
UGC07524	2	42.0	31	0	✓	✓	–
UGC07559	4	52.0	6	1	✓	✓	–
UGC07577	2	32.0	5	4	✓	✓	–
UGC07603	4	11.0	12	0	✓	✓	–
UGC07608	7	61.0	7	1	✓	✓	–
UGC07690	6	31.0	7	0	✓	✓	–
UGC07866	3	32.0	6	1	✓	✓	–
UGC08286	8	12.0	17	0	✓	✓	–
UGC08490	8	32.0	30	0	✓	✓	–
UGC08550	6	1.0	11	0	✓	✓	–
UGC08699	9	21.0	41	0	✓	✓	–
UGC08837	5	62.0	8	0	✓	✓	–
UGC09037	10	1.0	22	0	✓	✓	–
UGC09133	0	1.0	68	0	✓	✓	–
UGC09992	5	11.0	5	0	✓	✓	–
UGC10310	10	1.0	7	0	✓	✓	–

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Table S6: **Sample summary for the retained SPARC subset used in the matched comparisons (continued).**

Galaxy	Q	i [°]	$N_{\rm retained}$	$N_{\rm removed}$	Tier 1	Tier 2	Map pilot
UGC11455	3	1.0	36	0	✓	✓	–
UGC11557	7	51.0	10	2	✓	✓	–
UGC11820	3	31.0	10	0	✓	✓	–
UGC11914	9	1.0	65	0	✓	✓	–
UGC12506	9	1.0	31	0	✓	✓	–
UGC12632	3	31.0	15	0	✓	✓	–
UGC12732	7	1.0	16	0	✓	✓	–
UGCA281	2	82.0	6	1	✓	✓	–
UGCA442	4	22.0	8	0	✓	✓	–
UGCA444	2	52.0	35	1	✓	✓	–

S11.4 Per-galaxy matched-refit summary

The full per-galaxy matched-refit table is likewise given as a long table so that the per-model rows for each system can span multiple pages without splitting the header structure.

Table S7: **Per-galaxy matched-refit summary.** For each galaxy and each Tier 2 framework, the table reports the retained point count, the likelihood contribution χ^2, the reduced statistic χ^2_ν, and the MAP nuisance values actually used in the reported totals. The full table should be released in machine-readable form.

Galaxy	Model	$N_{\rm pts}$	χ^2	χ^2_ν	$s_D^{\rm MAP}$	$i^{\rm MAP}$	$\Upsilon_{\rm disc}^{\rm MAP}$	$\Upsilon_{\rm bul}^{\rm MAP}$	notes
CamB	VDII	8	19.869	2.484	0.640	42.500	0.509	N.A.	MAP nuisance fit
CamB	MOND	8	18.417	2.302	0.640	42.500	0.509	N.A.	MAP nuisance fit
CamB	NFW	8	11.292	1.411	0.760	55.500	0.741	N.A.	MAP nuisance + 2-parameter halo
CamB	Burkert	8	9.967	1.246	0.880	49.000	0.741	N.A.	MAP nuisance + 2-parameter halo
D512-2	VDII	4	0.203	0.051	0.898	61.000	0.799	N.A.	MAP nuisance fit
D512-2	MOND	4	0.633	0.158	0.898	61.000	1.000	N.A.	MAP nuisance fit
D512-2	NFW	4	0.113	0.028	1.000	61.000	1.000	N.A.	MAP nuisance + 2-parameter halo
D512-2	Burkert	4	0.033	0.008	1.000	61.000	1.000	N.A.	MAP nuisance + 2-parameter halo
D564-8	VDII	6	5.927	0.988	0.852	79.287	0.799	N.A.	MAP nuisance fit
D564-8	MOND	6	5.232	0.872	0.852	79.287	0.799	N.A.	MAP nuisance fit
D564-8	NFW	6	1.893	0.315	1.000	80.367	1.000	N.A.	MAP nuisance + 2-parameter halo
D564-8	Burkert	6	0.195	0.032	1.000	80.367	1.000	N.A.	MAP nuisance + 2-parameter halo
D631-7	VDII	16	34.808	2.175	0.846	79.737	0.638	N.A.	MAP nuisance fit
D631-7	MOND	16	34.427	2.152	0.846	79.737	0.638	N.A.	MAP nuisance fit
D631-7	NFW	16	21.437	1.340	0.897	80.767	0.741	N.A.	MAP nuisance + 2-parameter halo
D631-7	Burkert	16	5.670	0.354	1.000	80.767	0.741	N.A.	MAP nuisance + 2-parameter halo
DDO064	VDII	13	2.122	0.163	1.000	41.000	1.000	N.A.	MAP nuisance fit
DDO064	MOND	13	2.060	0.158	1.000	41.000	1.000	N.A.	MAP nuisance fit
DDO064	NFW	13	2.748	0.211	1.000	41.000	1.000	N.A.	MAP nuisance + 2-parameter halo
DDO064	Burkert	13	1.597	0.123	1.000	41.000	1.000	N.A.	MAP nuisance + 2-parameter halo
DDO154	VDII	12	7.292	0.608	1.000	64.000	0.799	N.A.	MAP nuisance fit
DDO154	MOND	12	7.476	0.623	1.000	64.000	0.799	N.A.	MAP nuisance fit
DDO154	NFW	12	3.023	0.252	1.000	64.000	0.741	N.A.	MAP nuisance + 2-parameter halo
DDO154	Burkert	12	0.713	0.059	1.000	64.000	1.000	N.A.	MAP nuisance + 2-parameter halo
DDO161	VDII	30	7.782	0.259	0.892	45.750	0.638	N.A.	MAP nuisance fit
DDO161	MOND	30	8.863	0.295	0.946	45.750	0.638	N.A.	MAP nuisance fit
DDO161	NFW	30	4.574	0.152	1.000	51.000	0.741	N.A.	MAP nuisance + 2-parameter halo
DDO161	Burkert	30	1.022	0.034	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
DDO168	VDII	10	6.758	0.676	1.000	11.475	1.000	N.A.	MAP nuisance fit
DDO168	MOND	10	6.663	0.666	1.000	11.475	1.000	N.A.	MAP nuisance fit
DDO168	NFW	10	5.965	0.597	1.000	13.633	1.000	N.A.	MAP nuisance + 2-parameter halo
DDO168	Burkert	10	2.245	0.224	1.000	13.633	1.000	N.A.	MAP nuisance + 2-parameter halo
DDO170	VDII	8	0.801	0.100	1.000	18.425	1.000	N.A.	MAP nuisance fit
DDO170	MOND	8	0.922	0.115	1.000	18.425	1.000	N.A.	MAP nuisance fit
DDO170	NFW	8	0.751	0.094	1.000	22.900	1.000	N.A.	MAP nuisance + 2-parameter halo

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Table S7: **Per-galaxy matched-refit summary (continued).**

Galaxy	Model	$N_{\rm pts}$	χ^2	χ^2_ν	$s_D^{\rm MAP}$	$i^{\rm MAP}$	$\Upsilon_{\rm disc}^{\rm MAP}$	$\Upsilon_{\rm bul}^{\rm MAP}$	notes
DDO170	Burkert	8	0.234	0.029	1.000	22.900	1.000	N.A.	MAP nuisance + 2-parameter halo
ESO079-G014	VDII	14	5.152	0.368	0.876	73.362	1.000	N.A.	MAP nuisance fit
ESO079-G014	MOND	14	5.275	0.377	0.876	68.150	1.000	N.A.	MAP nuisance fit
ESO079-G014	NFW	14	5.093	0.364	1.000	68.150	1.000	N.A.	MAP nuisance + 2-parameter halo
ESO079-G014	Burkert	14	2.550	0.182	1.000	68.150	1.000	N.A.	MAP nuisance + 2-parameter halo
ESO116-G012	VDII	15	5.108	0.341	1.000	74.000	1.000	N.A.	MAP nuisance fit
ESO116-G012	MOND	15	5.543	0.370	1.000	74.000	1.000	N.A.	MAP nuisance fit
ESO116-G012	NFW	15	2.458	0.164	1.000	74.000	1.000	N.A.	MAP nuisance + 2-parameter halo
ESO116-G012	Burkert	15	0.912	0.061	1.000	74.000	1.000	N.A.	MAP nuisance + 2-parameter halo
ESO444-G084	VDII	7	4.908	0.701	1.250	82.775	1.252	N.A.	MAP nuisance fit
ESO444-G084	MOND	7	5.436	0.777	1.250	82.775	1.252	N.A.	MAP nuisance fit
ESO444-G084	NFW	7	0.531	0.076	1.000	80.700	1.000	N.A.	MAP nuisance + 2-parameter halo
ESO444-G084	Burkert	7	0.511	0.073	1.000	80.700	1.000	N.A.	MAP nuisance + 2-parameter halo
ESO563-G021	VDII	29	40.391	1.393	0.875	83.000	1.000	N.A.	MAP nuisance fit
ESO563-G021	MOND	29	42.265	1.457	0.875	83.000	1.000	N.A.	MAP nuisance fit
ESO563-G021	NFW	29	32.861	1.133	1.000	83.000	0.741	N.A.	MAP nuisance + 2-parameter halo
ESO563-G021	Burkert	29	13.984	0.482	0.833	83.000	0.549	N.A.	MAP nuisance + 2-parameter halo
F561-1	VDII	5	11.091	2.218	0.969	22.500	0.509	N.A.	MAP nuisance fit
F561-1	MOND	5	9.988	1.998	0.969	22.500	0.509	N.A.	MAP nuisance fit
F561-1	NFW	5	1.026	0.205	1.000	24.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F561-1	Burkert	5	0.689	0.138	1.000	24.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F563-1	VDII	17	11.654	0.686	1.000	25.750	1.568	N.A.	MAP nuisance fit
F563-1	MOND	17	12.479	0.734	1.000	25.750	1.568	N.A.	MAP nuisance fit
F563-1	NFW	17	7.283	0.428	1.000	25.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F563-1	Burkert	17	3.472	0.204	1.000	25.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F563-V1	VDII	5	33.932	6.786	0.986	59.250	0.407	N.A.	MAP nuisance fit
F563-V1	MOND	5	31.345	6.269	0.986	59.250	0.407	N.A.	MAP nuisance fit
F563-V1	NFW	5	1.227	0.245	1.000	60.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F563-V1	Burkert	5	0.367	0.073	1.000	60.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F563-V2	VDII	7	3.134	0.448	1.089	29.750	1.568	N.A.	MAP nuisance fit
F563-V2	MOND	7	3.330	0.476	1.089	29.750	1.568	N.A.	MAP nuisance fit
F563-V2	NFW	7	1.901	0.272	1.000	29.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F563-V2	Burkert	7	0.441	0.063	1.000	29.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F565-V2	VDII	6	1.916	0.319	1.118	60.000	1.252	N.A.	MAP nuisance fit
F565-V2	MOND	6	2.277	0.379	1.118	60.000	1.252	N.A.	MAP nuisance fit
F565-V2	NFW	6	0.322	0.054	1.000	60.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F565-V2	Burkert	6	0.085	0.014	1.000	60.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F567-2	VDII	5	3.675	0.735	0.916	19.250	0.799	N.A.	MAP nuisance fit
F567-2	MOND	5	3.174	0.635	0.916	19.250	0.799	N.A.	MAP nuisance fit
F567-2	NFW	5	0.544	0.109	1.000	20.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F567-2	Burkert	5	0.353	0.071	1.000	20.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F568-1	VDII	11	5.385	0.490	1.090	26.750	1.568	N.A.	MAP nuisance fit
F568-1	MOND	11	5.764	0.524	1.090	26.750	1.568	N.A.	MAP nuisance fit
F568-1	NFW	11	1.870	0.170	1.000	26.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F568-1	Burkert	11	0.378	0.034	1.000	26.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F568-3	VDII	17	7.751	0.456	1.000	41.000	1.000	N.A.	MAP nuisance fit
F568-3	MOND	17	7.591	0.447	1.000	41.000	1.000	N.A.	MAP nuisance fit
F568-3	NFW	17	7.581	0.446	1.000	41.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F568-3	Burkert	17	2.962	0.174	1.000	41.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F568-V1	VDII	13	1.857	0.143	1.058	64.000	1.964	N.A.	MAP nuisance fit
F568-V1	MOND	13	2.083	0.160	1.058	64.000	1.964	N.A.	MAP nuisance fit
F568-V1	NFW	13	0.539	0.041	1.000	61.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F568-V1	Burkert	13	0.179	0.014	1.000	61.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F571-8	VDII	12	35.780	2.982	0.959	85.000	0.509	N.A.	MAP nuisance fit
F571-8	MOND	12	37.764	3.147	0.959	85.000	0.509	N.A.	MAP nuisance fit
F571-8	NFW	12	15.587	1.299	0.946	85.000	0.407	N.A.	MAP nuisance + 2-parameter halo
F571-8	Burkert	12	7.339	0.612	0.946	85.000	0.407	N.A.	MAP nuisance + 2-parameter halo
F571-V1	VDII	7	0.870	0.124	1.000	30.000	1.000	N.A.	MAP nuisance fit

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Table S7: **Per-galaxy matched-refit summary (continued).**

Galaxy	Model	$N_{\rm pts}$	χ^2	χ^2_ν	$s_D^{\rm MAP}$	$i^{\rm MAP}$	$\Upsilon_{\rm disc}^{\rm MAP}$	$\Upsilon_{\rm bul}^{\rm MAP}$	notes
F571-V1	MOND	7	0.906	0.129	1.000	30.000	1.000	N.A.	MAP nuisance fit
F571-V1	NFW	7	0.646	0.092	1.000	30.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F571-V1	Burkert	7	0.113	0.016	1.000	30.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F574-1	VDII	14	4.196	0.300	1.000	82.125	1.000	N.A.	MAP nuisance fit
F574-1	MOND	14	3.752	0.268	1.076	82.125	1.000	N.A.	MAP nuisance fit
F574-1	NFW	14	4.054	0.290	1.000	79.833	1.000	N.A.	MAP nuisance + 2-parameter halo
F574-1	Burkert	14	0.463	0.033	1.000	79.833	1.000	N.A.	MAP nuisance + 2-parameter halo
F574-2	VDII	3	0.033	0.011	1.000	5.000	1.000	N.A.	MAP nuisance fit
F574-2	MOND	3	0.056	0.019	1.000	5.000	1.000	N.A.	MAP nuisance fit
F574-2	NFW	3	0.684	0.228	1.000	10.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F574-2	Burkert	3	0.225	0.075	1.000	10.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F579-V1	VDII	14	5.900	0.421	1.000	51.000	1.252	N.A.	MAP nuisance fit
F579-V1	MOND	14	5.859	0.418	1.000	51.000	1.252	N.A.	MAP nuisance fit
F579-V1	NFW	14	0.760	0.054	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F579-V1	Burkert	14	0.444	0.032	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F583-1	VDII	20	4.846	0.242	1.086	55.725	1.000	N.A.	MAP nuisance fit
F583-1	MOND	20	4.581	0.229	1.000	55.725	1.252	N.A.	MAP nuisance fit
F583-1	NFW	20	3.586	0.179	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F583-1	Burkert	20	0.459	0.023	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F583-4	VDII	11	0.422	0.038	1.000	55.000	0.799	N.A.	MAP nuisance fit
F583-4	MOND	11	0.300	0.027	1.000	55.000	0.799	N.A.	MAP nuisance fit
F583-4	NFW	11	0.205	0.019	1.000	55.000	1.000	N.A.	MAP nuisance + 2-parameter halo
F583-4	Burkert	11	0.622	0.057	1.000	55.000	1.000	N.A.	MAP nuisance + 2-parameter halo
IC2574	VDII	34	17.155	0.505	0.625	75.000	0.799	N.A.	MAP nuisance fit
IC2574	MOND	34	17.393	0.512	0.625	75.000	0.799	N.A.	MAP nuisance fit
IC2574	NFW	34	21.439	0.631	1.000	75.000	1.000	N.A.	MAP nuisance + 2-parameter halo
IC2574	Burkert	34	2.196	0.065	1.000	75.000	1.000	N.A.	MAP nuisance + 2-parameter halo
IC4202	VDII	32	92.547	2.892	1.000	89.000	0.407	N.A.	MAP nuisance fit
IC4202	MOND	32	95.868	2.996	1.000	89.000	0.407	N.A.	MAP nuisance fit
IC4202	NFW	32	74.501	2.328	0.960	89.000	0.407	N.A.	MAP nuisance + 2-parameter halo
IC4202	Burkert	32	54.824	1.713	0.960	89.000	0.407	N.A.	MAP nuisance + 2-parameter halo
KK98-251	VDII	14	3.467	0.248	0.900	32.150	1.000	N.A.	MAP nuisance fit
KK98-251	MOND	14	3.529	0.252	0.799	36.575	0.799	N.A.	MAP nuisance fit
KK98-251	NFW	14	4.307	0.308	1.000	41.000	1.000	N.A.	MAP nuisance + 2-parameter halo
KK98-251	Burkert	14	0.454	0.032	1.000	41.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC0024	VDII	28	4.129	0.147	1.039	66.800	1.252	N.A.	MAP nuisance fit
NGC0024	MOND	28	4.024	0.144	1.039	66.800	1.252	N.A.	MAP nuisance fit
NGC0024	NFW	28	2.975	0.106	1.000	62.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC0024	Burkert	28	1.905	0.068	1.000	62.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC0055	VDII	21	6.427	0.306	0.961	12.525	0.509	N.A.	MAP nuisance fit
NGC0055	MOND	21	8.467	0.403	1.039	8.762	1.252	N.A.	MAP nuisance fit
NGC0055	NFW	21	3.509	0.167	1.000	10.017	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC0055	Burkert	21	0.725	0.035	1.000	10.017	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC0100	VDII	20	10.086	0.504	1.000	44.325	1.000	N.A.	MAP nuisance fit
NGC0100	MOND	20	10.529	0.526	1.000	44.325	1.000	N.A.	MAP nuisance fit
NGC0100	NFW	20	4.830	0.241	1.000	51.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC0100	Burkert	20	1.570	0.078	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC0247	VDII	26	1.196	0.046	1.000	74.000	1.000	N.A.	MAP nuisance fit
NGC0247	MOND	26	1.183	0.045	1.000	74.000	1.000	N.A.	MAP nuisance fit
NGC0247	NFW	26	0.973	0.037	1.000	74.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC0247	Burkert	26	2.989	0.115	1.000	74.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC0289	VDII	28	3.104	0.111	0.971	46.000	0.799	N.A.	MAP nuisance fit
NGC0289	MOND	28	3.484	0.124	1.000	45.250	0.799	N.A.	MAP nuisance fit
NGC0289	NFW	28	2.639	0.094	1.000	46.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC0289	Burkert	28	1.847	0.066	1.000	46.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC0300	VDII	25	4.596	0.184	1.000	43.500	1.000	N.A.	MAP nuisance fit
NGC0300	MOND	25	5.433	0.217	1.029	42.000	1.000	N.A.	MAP nuisance fit
NGC0300	NFW	25	2.117	0.085	1.000	42.000	1.000	N.A.	MAP nuisance + 2-parameter halo

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Table S7: **Per-galaxy matched-refit summary (continued).**

Galaxy	Model	$N_{\rm pts}$	χ^2	χ^2_ν	$s_D^{\rm MAP}$	$i^{\rm MAP}$	$\Upsilon_{\rm disc}^{\rm MAP}$	$\Upsilon_{\rm bul}^{\rm MAP}$	notes
NGC0300	Burkert	25	0.857	0.034	1.000	42.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC0801	VDII	13	2.405	0.185	0.817	57.500	1.000	N.A.	MAP nuisance fit
NGC0801	MOND	13	2.751	0.212	0.817	57.500	1.000	N.A.	MAP nuisance fit
NGC0801	NFW	13	3.573	0.275	1.000	68.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC0801	Burkert	13	3.508	0.270	0.756	68.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC0891	VDII	18	7.986	0.444	0.931	89.000	0.638	N.A.	MAP nuisance fit
NGC0891	MOND	18	7.469	0.415	0.862	89.000	0.638	N.A.	MAP nuisance fit
NGC0891	NFW	18	9.246	0.514	0.816	89.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC0891	Burkert	18	16.132	0.896	0.908	89.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC1003	VDII	36	18.046	0.501	1.000	18.538	1.252	N.A.	MAP nuisance fit
NGC1003	MOND	36	8.961	0.249	1.000	23.050	0.638	N.A.	MAP nuisance fit
NGC1003	NFW	36	3.890	0.108	1.000	23.050	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC1003	Burkert	36	1.668	0.046	0.934	23.050	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC1090	VDII	23	5.370	0.233	1.000	41.400	1.000	N.A.	MAP nuisance fit
NGC1090	MOND	23	6.162	0.268	1.000	41.400	1.000	N.A.	MAP nuisance fit
NGC1090	NFW	23	7.086	0.308	1.000	44.600	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC1090	Burkert	23	5.761	0.250	1.000	44.600	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC1705	VDII	14	3.386	0.242	1.191	80.000	1.568	N.A.	MAP nuisance fit
NGC1705	MOND	14	3.993	0.285	1.191	80.000	1.568	N.A.	MAP nuisance fit
NGC1705	NFW	14	0.259	0.019	1.000	81.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC1705	Burkert	14	1.201	0.086	1.000	81.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC2366	VDII	25	9.077	0.363	0.932	56.900	0.509	N.A.	MAP nuisance fit
NGC2366	MOND	25	11.612	0.464	1.000	51.800	0.638	N.A.	MAP nuisance fit
NGC2366	NFW	25	12.551	0.502	1.000	62.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC2366	Burkert	25	3.899	0.156	1.000	62.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC2403	VDII	73	25.087	0.344	1.000	57.275	1.000	N.A.	MAP nuisance fit
NGC2403	MOND	73	29.260	0.401	1.000	57.275	1.000	N.A.	MAP nuisance fit
NGC2403	NFW	73	4.903	0.067	1.000	68.300	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC2403	Burkert	73	6.612	0.091	1.000	62.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC2683	VDII	11	1.116	0.101	1.000	80.000	0.799	N.A.	MAP nuisance fit
NGC2683	MOND	11	1.154	0.105	1.000	80.000	0.799	N.A.	MAP nuisance fit
NGC2683	NFW	11	1.984	0.180	1.000	81.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC2683	Burkert	11	1.758	0.160	1.000	81.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC2841	VDII	50	9.720	0.194	1.107	78.250	1.000	N.A.	MAP nuisance fit
NGC2841	MOND	50	10.793	0.216	1.107	78.250	1.000	N.A.	MAP nuisance fit
NGC2841	NFW	50	0.536	0.011	1.000	76.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC2841	Burkert	50	1.111	0.022	1.000	76.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC2903	VDII	34	93.746	2.757	0.902	75.100	0.407	N.A.	MAP nuisance fit
NGC2903	MOND	34	96.213	2.830	0.902	75.100	0.407	N.A.	MAP nuisance fit
NGC2903	NFW	34	50.505	1.485	0.739	70.467	0.407	N.A.	MAP nuisance + 2-parameter halo
NGC2903	Burkert	34	32.030	0.942	0.608	70.467	0.407	N.A.	MAP nuisance + 2-parameter halo
NGC2915	VDII	27	13.061	0.484	1.333	57.500	1.568	N.A.	MAP nuisance fit
NGC2915	MOND	27	14.724	0.545	1.333	57.500	1.568	N.A.	MAP nuisance fit
NGC2915	NFW	27	4.075	0.151	1.000	56.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC2915	Burkert	27	2.110	0.078	1.000	56.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC2955	VDII	24	3.712	0.155	0.930	55.250	0.799	N.A.	MAP nuisance fit
NGC2955	MOND	24	3.130	0.130	0.861	56.000	0.799	N.A.	MAP nuisance fit
NGC2955	NFW	24	6.253	0.261	0.814	56.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC2955	Burkert	24	5.385	0.224	0.814	56.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC2976	VDII	25	5.072	0.203	0.836	79.513	0.799	N.A.	MAP nuisance fit
NGC2976	MOND	25	4.972	0.199	0.836	79.513	0.799	N.A.	MAP nuisance fit
NGC2976	NFW	25	2.797	0.112	0.891	80.567	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC2976	Burkert	25	2.774	0.111	0.891	80.567	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC2998	VDII	13	5.365	0.413	0.875	58.000	1.000	N.A.	MAP nuisance fit
NGC2998	MOND	13	5.084	0.391	0.875	58.000	1.000	N.A.	MAP nuisance fit
NGC2998	NFW	13	4.425	0.340	1.000	58.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC2998	Burkert	13	5.003	0.385	1.000	58.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3109	VDII	25	12.718	0.509	1.250	74.750	0.799	N.A.	MAP nuisance fit

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Table S7: **Per-galaxy matched-refit summary (continued).**

Galaxy	Model	$N_{\rm pts}$	χ^2	χ^2_ν	$s_D^{\rm MAP}$	$i^{\rm MAP}$	$\Upsilon_{\rm disc}^{\rm MAP}$	$\Upsilon_{\rm bul}^{\rm MAP}$	notes
NGC3109	MOND	25	12.685	0.507	1.375	70.000	0.799	N.A.	MAP nuisance fit
NGC3109	NFW	25	23.026	0.921	1.000	70.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3109	Burkert	25	0.541	0.022	1.000	76.333	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3198	VDII	40	4.113	0.103	1.000	73.000	0.799	N.A.	MAP nuisance fit
NGC3198	MOND	40	4.829	0.121	1.000	73.000	0.799	N.A.	MAP nuisance fit
NGC3198	NFW	40	6.808	0.170	1.000	73.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3198	Burkert	40	4.075	0.102	1.000	73.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3521	VDII	41	5.613	0.137	0.664	75.000	1.000	N.A.	MAP nuisance fit
NGC3521	MOND	41	6.941	0.169	0.664	75.000	1.000	N.A.	MAP nuisance fit
NGC3521	NFW	41	3.071	0.075	0.851	75.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC3521	Burkert	41	5.202	0.127	0.701	75.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3726	VDII	12	2.246	0.187	0.917	50.000	0.638	N.A.	MAP nuisance fit
NGC3726	MOND	12	2.580	0.215	0.917	50.000	0.638	N.A.	MAP nuisance fit
NGC3726	NFW	12	2.115	0.176	1.000	49.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC3726	Burkert	12	2.147	0.179	0.778	53.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3741	VDII	21	11.459	0.546	1.052	79.500	0.799	N.A.	MAP nuisance fit
NGC3741	MOND	21	12.744	0.607	1.052	79.500	0.799	N.A.	MAP nuisance fit
NGC3741	NFW	21	1.664	0.079	1.000	70.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3741	Burkert	21	1.732	0.082	1.000	70.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3769	VDII	12	3.776	0.315	0.917	70.000	0.799	N.A.	MAP nuisance fit
NGC3769	MOND	12	4.376	0.365	0.917	70.000	0.799	N.A.	MAP nuisance fit
NGC3769	NFW	12	3.551	0.296	1.000	70.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC3769	Burkert	12	2.487	0.207	1.000	70.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC3877	VDII	13	17.161	1.320	0.750	73.000	0.799	N.A.	MAP nuisance fit
NGC3877	MOND	13	17.990	1.384	0.917	73.000	0.638	N.A.	MAP nuisance fit
NGC3877	NFW	13	19.362	1.489	0.889	76.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC3877	Burkert	13	19.764	1.520	0.889	76.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC3893	VDII	10	2.960	0.296	0.917	46.000	1.000	N.A.	MAP nuisance fit
NGC3893	MOND	10	2.976	0.298	1.000	49.000	0.799	N.A.	MAP nuisance fit
NGC3893	NFW	10	1.129	0.113	1.000	45.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC3893	Burkert	10	1.858	0.186	0.889	45.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3917	VDII	17	9.333	0.549	1.000	78.000	0.799	N.A.	MAP nuisance fit
NGC3917	MOND	17	9.303	0.547	1.000	78.000	0.799	N.A.	MAP nuisance fit
NGC3917	NFW	17	9.400	0.553	1.000	78.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3917	Burkert	17	5.387	0.317	1.000	78.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3949	VDII	7	1.283	0.183	0.917	52.000	0.799	N.A.	MAP nuisance fit
NGC3949	MOND	7	1.733	0.248	0.917	52.000	0.799	N.A.	MAP nuisance fit
NGC3949	NFW	7	2.624	0.375	0.889	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3949	Burkert	7	2.602	0.372	0.889	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3953	VDII	8	0.415	0.052	1.000	59.000	0.799	N.A.	MAP nuisance fit
NGC3953	MOND	8	0.613	0.077	1.000	59.000	0.799	N.A.	MAP nuisance fit
NGC3953	NFW	8	0.239	0.030	0.889	62.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3953	Burkert	8	0.225	0.028	0.889	62.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3972	VDII	10	2.815	0.282	0.917	77.000	1.000	N.A.	MAP nuisance fit
NGC3972	MOND	10	2.938	0.294	0.917	77.000	1.000	N.A.	MAP nuisance fit
NGC3972	NFW	10	2.844	0.284	1.000	77.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3972	Burkert	10	1.761	0.176	1.000	77.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3992	VDII	9	0.205	0.023	1.000	56.000	1.000	N.A.	MAP nuisance fit
NGC3992	MOND	9	0.230	0.026	1.000	56.000	1.000	N.A.	MAP nuisance fit
NGC3992	NFW	9	0.100	0.011	1.000	56.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC3992	Burkert	9	0.164	0.018	1.000	56.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4010	VDII	12	4.697	0.391	0.917	89.000	0.799	N.A.	MAP nuisance fit
NGC4010	MOND	12	4.886	0.407	0.917	89.000	0.799	N.A.	MAP nuisance fit
NGC4010	NFW	12	6.222	0.519	1.000	89.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4010	Burkert	12	4.495	0.375	1.000	89.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4013	VDII	36	6.098	0.169	0.833	89.000	1.000	N.A.	MAP nuisance fit
NGC4013	MOND	36	7.425	0.206	0.833	89.000	1.000	N.A.	MAP nuisance fit
NGC4013	NFW	36	5.020	0.139	0.889	89.000	1.000	N.A.	MAP nuisance + 2-parameter halo

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Table S7: **Per-galaxy matched-refit summary (continued).**

Galaxy	Model	$N_{\rm pts}$	χ^2	χ^2_ν	$s_D^{\rm MAP}$	$i^{\rm MAP}$	$\Upsilon_{\rm disc}^{\rm MAP}$	$\Upsilon_{\rm bul}^{\rm MAP}$	notes
NGC4013	Burkert	36	5.659	0.157	1.000	89.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4051	VDII	7	1.000	0.143	1.000	46.000	0.638	N.A.	MAP nuisance fit
NGC4051	MOND	7	0.514	0.073	0.917	46.000	0.638	N.A.	MAP nuisance fit
NGC4051	NFW	7	1.416	0.202	0.889	45.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4051	Burkert	7	1.377	0.197	0.889	45.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4068	VDII	5	3.993	0.799	0.931	18.700	0.799	N.A.	MAP nuisance fit
NGC4068	MOND	5	3.718	0.744	0.931	18.700	0.799	N.A.	MAP nuisance fit
NGC4068	NFW	5	3.013	0.603	1.000	22.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4068	Burkert	5	1.885	0.377	1.000	22.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4085	VDII	7	8.075	1.154	0.917	81.875	0.509	N.A.	MAP nuisance fit
NGC4085	MOND	7	8.650	1.236	0.917	81.875	0.509	N.A.	MAP nuisance fit
NGC4085	NFW	7	8.161	1.166	0.889	82.667	0.549	N.A.	MAP nuisance + 2-parameter halo
NGC4085	Burkert	7	9.553	1.365	1.000	82.667	0.549	N.A.	MAP nuisance + 2-parameter halo
NGC4088	VDII	12	2.693	0.224	0.833	66.000	0.638	N.A.	MAP nuisance fit
NGC4088	MOND	12	3.308	0.276	0.833	66.000	0.638	N.A.	MAP nuisance fit
NGC4088	NFW	12	4.945	0.412	0.889	65.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC4088	Burkert	12	6.809	0.567	0.889	69.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC4100	VDII	24	1.197	0.050	0.917	73.000	1.000	N.A.	MAP nuisance fit
NGC4100	MOND	24	1.471	0.061	0.917	73.000	1.000	N.A.	MAP nuisance fit
NGC4100	NFW	24	1.667	0.069	1.000	73.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4100	Burkert	24	1.398	0.058	1.000	73.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4138	VDII	7	0.793	0.113	1.000	53.000	0.799	N.A.	MAP nuisance fit
NGC4138	MOND	7	1.174	0.168	0.917	50.000	1.000	N.A.	MAP nuisance fit
NGC4138	NFW	7	2.104	0.301	1.000	53.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4138	Burkert	7	1.919	0.274	1.000	53.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4157	VDII	17	4.405	0.259	0.833	81.875	0.799	N.A.	MAP nuisance fit
NGC4157	MOND	17	3.362	0.198	0.917	81.875	0.638	N.A.	MAP nuisance fit
NGC4157	NFW	17	3.508	0.206	0.889	82.667	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC4157	Burkert	17	2.134	0.126	0.889	82.667	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC4183	VDII	23	1.007	0.044	0.917	81.875	1.000	N.A.	MAP nuisance fit
NGC4183	MOND	23	1.369	0.060	1.000	81.875	1.000	N.A.	MAP nuisance fit
NGC4183	NFW	23	0.388	0.017	1.000	82.667	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4183	Burkert	23	1.088	0.047	1.000	82.667	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4214	VDII	14	4.015	0.287	1.375	42.000	1.000	N.A.	MAP nuisance fit
NGC4214	MOND	14	4.435	0.317	1.375	42.000	1.000	N.A.	MAP nuisance fit
NGC4214	NFW	14	2.114	0.151	1.000	42.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4214	Burkert	14	2.900	0.207	1.000	42.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC4217	VDII	19	34.325	1.807	0.833	85.250	0.509	N.A.	MAP nuisance fit
NGC4217	MOND	19	32.457	1.708	0.917	85.250	0.407	N.A.	MAP nuisance fit
NGC4217	NFW	19	19.538	1.028	0.778	86.500	0.407	N.A.	MAP nuisance + 2-parameter halo
NGC4217	Burkert	19	10.389	0.547	0.667	86.500	0.407	N.A.	MAP nuisance + 2-parameter halo
NGC4389	VDII	6	11.241	1.874	0.833	38.000	0.509	N.A.	MAP nuisance fit
NGC4389	MOND	6	11.923	1.987	0.833	38.000	0.509	N.A.	MAP nuisance fit
NGC4389	NFW	6	10.584	1.764	0.889	46.000	0.407	N.A.	MAP nuisance + 2-parameter halo
NGC4389	Burkert	6	8.589	1.431	0.889	42.000	0.407	N.A.	MAP nuisance + 2-parameter halo
NGC4559	VDII	32	2.878	0.090	0.978	67.000	0.638	N.A.	MAP nuisance fit
NGC4559	MOND	32	2.925	0.091	0.978	67.000	0.638	N.A.	MAP nuisance fit
NGC4559	NFW	32	2.468	0.077	1.000	67.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC4559	Burkert	32	1.068	0.033	1.000	67.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC5005	VDII	18	0.984	0.055	1.000	68.000	0.638	N.A.	MAP nuisance fit
NGC5005	MOND	18	1.356	0.075	1.000	68.000	0.638	N.A.	MAP nuisance fit
NGC5005	NFW	18	0.296	0.016	1.000	68.000	0.549	N.A.	MAP nuisance + 2-parameter halo
NGC5005	Burkert	18	0.858	0.048	0.947	63.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC5033	VDII	22	6.886	0.313	1.000	66.000	0.799	N.A.	MAP nuisance fit
NGC5033	MOND	22	7.981	0.363	1.000	66.000	0.799	N.A.	MAP nuisance fit
NGC5033	NFW	22	4.831	0.220	1.000	66.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC5033	Burkert	22	3.652	0.166	1.000	66.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC5055	VDII	28	4.273	0.153	0.932	53.500	0.638	N.A.	MAP nuisance fit

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Table S7: **Per-galaxy matched-refit summary (continued).**

Galaxy	Model	$N_{\rm pts}$	χ^2	χ^2_ν	$s_D^{\rm MAP}$	$i^{\rm MAP}$	$\Upsilon_{\rm disc}^{\rm MAP}$	$\Upsilon_{\rm bul}^{\rm MAP}$	notes
NGC5055	MOND	28	4.054	0.145	0.864	53.500	0.638	N.A.	MAP nuisance fit
NGC5055	NFW	28	4.799	0.171	0.818	53.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC5055	Burkert	28	3.923	0.140	0.818	53.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC5371	VDII	19	3.095	0.163	0.637	21.000	1.000	N.A.	MAP nuisance fit
NGC5371	MOND	19	3.118	0.164	0.637	21.000	1.000	N.A.	MAP nuisance fit
NGC5371	NFW	19	3.969	0.209	0.838	21.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC5371	Burkert	19	4.071	0.214	0.838	21.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC5585	VDII	24	16.872	0.703	1.060	17.175	0.799	N.A.	MAP nuisance fit
NGC5585	MOND	24	18.019	0.751	1.060	17.175	0.799	N.A.	MAP nuisance fit
NGC5585	NFW	24	9.505	0.396	0.919	21.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC5585	Burkert	24	4.235	0.176	1.000	21.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC5907	VDII	19	1.067	0.056	0.875	88.125	1.000	N.A.	MAP nuisance fit
NGC5907	MOND	19	1.032	0.054	0.875	88.125	1.000	N.A.	MAP nuisance fit
NGC5907	NFW	19	0.754	0.040	1.000	87.833	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC5907	Burkert	19	0.764	0.040	1.000	87.833	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC5985	VDII	33	4.198	0.127	1.000	60.000	1.568	N.A.	MAP nuisance fit
NGC5985	MOND	33	4.093	0.124	1.000	60.000	1.568	N.A.	MAP nuisance fit
NGC5985	NFW	33	0.958	0.029	1.000	60.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC5985	Burkert	33	1.019	0.031	1.000	60.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC6015	VDII	44	7.089	0.161	1.000	60.000	1.000	N.A.	MAP nuisance fit
NGC6015	MOND	44	8.293	0.188	1.000	60.000	1.000	N.A.	MAP nuisance fit
NGC6015	NFW	44	4.896	0.111	1.000	60.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC6015	Burkert	44	4.297	0.098	1.000	60.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC6195	VDII	23	1.399	0.061	0.953	62.000	0.638	N.A.	MAP nuisance fit
NGC6195	MOND	23	1.669	0.073	0.953	62.000	0.638	N.A.	MAP nuisance fit
NGC6195	NFW	23	3.041	0.132	1.000	62.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC6195	Burkert	23	3.854	0.168	1.000	62.000	0.741	N.A.	MAP nuisance + 2-parameter halo
NGC6503	VDII	31	4.823	0.156	1.000	12.300	0.799	N.A.	MAP nuisance fit
NGC6503	MOND	31	6.166	0.199	1.000	12.300	0.799	N.A.	MAP nuisance fit
NGC6503	NFW	31	1.995	0.064	1.000	9.867	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC6503	Burkert	31	1.677	0.054	1.000	9.867	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC6674	VDII	14	0.911	0.065	1.000	54.000	1.000	N.A.	MAP nuisance fit
NGC6674	MOND	14	0.983	0.070	1.000	54.000	1.000	N.A.	MAP nuisance fit
NGC6674	NFW	14	0.396	0.028	1.000	54.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC6674	Burkert	14	0.825	0.059	1.000	54.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC6789	VDII	4	3.617	0.904	1.111	81.537	1.252	N.A.	MAP nuisance fit
NGC6789	MOND	4	3.672	0.918	1.111	81.537	1.252	N.A.	MAP nuisance fit
NGC6789	NFW	4	1.622	0.406	1.000	82.367	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC6789	Burkert	4	0.271	0.068	1.000	82.367	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC6946	VDII	58	5.459	0.094	0.820	53.450	1.000	N.A.	MAP nuisance fit
NGC6946	MOND	58	6.722	0.116	0.760	56.300	1.000	N.A.	MAP nuisance fit
NGC6946	NFW	58	8.314	0.143	0.840	54.400	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC6946	Burkert	58	6.073	0.105	0.840	54.400	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC7331	VDII	36	9.497	0.264	0.964	72.750	0.638	N.A.	MAP nuisance fit
NGC7331	MOND	36	7.101	0.197	1.000	75.000	0.509	N.A.	MAP nuisance fit
NGC7331	NFW	36	3.991	0.111	0.952	75.000	0.549	N.A.	MAP nuisance + 2-parameter halo
NGC7331	Burkert	36	3.066	0.085	0.952	72.000	0.549	N.A.	MAP nuisance + 2-parameter halo
NGC7793	VDII	43	5.764	0.134	0.750	83.725	1.000	N.A.	MAP nuisance fit
NGC7793	MOND	43	5.993	0.139	0.750	83.725	1.000	N.A.	MAP nuisance fit
NGC7793	NFW	43	9.464	0.220	1.000	78.450	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC7793	Burkert	43	6.879	0.160	1.000	78.450	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC7814	VDII	18	7.781	0.432	1.000	62.000	1.000	N.A.	MAP nuisance fit
NGC7814	MOND	18	8.482	0.471	1.000	62.000	1.000	N.A.	MAP nuisance fit
NGC7814	NFW	18	1.970	0.109	1.000	53.000	1.000	N.A.	MAP nuisance + 2-parameter halo
NGC7814	Burkert	18	1.592	0.088	1.000	53.000	1.000	N.A.	MAP nuisance + 2-parameter halo
PGC51017	VDII	6	42.245	7.041	0.840	63.000	0.407	N.A.	MAP nuisance fit
PGC51017	MOND	6	39.177	6.530	0.840	63.000	0.407	N.A.	MAP nuisance fit
PGC51017	NFW	6	5.118	0.853	1.000	66.000	0.741	N.A.	MAP nuisance + 2-parameter halo

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Table S7: **Per-galaxy matched-refit summary (continued).**

Galaxy	Model	$N_{\rm pts}$	χ^2	χ^2_ν	$s_D^{\rm MAP}$	$i^{\rm MAP}$	$\Upsilon^{\rm MAP}_{\rm disc}$	$\Upsilon^{\rm MAP}_{\rm bul}$	notes
PGC51017	Burkert	6	2.078	0.346	1.000	66.000	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC00128	VDII	21	1.330	0.063	1.000	57.000	1.252	N.A.	MAP nuisance fit
UGC00128	MOND	21	1.097	0.052	1.044	57.000	1.252	N.A.	MAP nuisance fit
UGC00128	NFW	21	0.297	0.014	1.000	57.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC00128	Burkert	21	1.038	0.049	1.000	57.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC00191	VDII	9	1.530	0.170	1.000	45.000	1.000	N.A.	MAP nuisance fit
UGC00191	MOND	9	1.606	0.178	1.000	45.000	1.000	N.A.	MAP nuisance fit
UGC00191	NFW	9	1.562	0.174	1.000	45.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC00191	Burkert	9	0.676	0.075	1.000	45.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC00634	VDII	4	1.055	0.264	1.000	37.000	1.000	N.A.	MAP nuisance fit
UGC00634	MOND	4	1.270	0.317	1.000	37.000	1.000	N.A.	MAP nuisance fit
UGC00634	NFW	4	0.217	0.054	1.000	37.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC00634	Burkert	4	0.030	0.008	1.000	37.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC00731	VDII	12	7.150	0.596	1.000	55.275	1.568	N.A.	MAP nuisance fit
UGC00731	MOND	12	7.175	0.598	1.031	55.275	1.568	N.A.	MAP nuisance fit
UGC00731	NFW	12	0.264	0.022	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC00731	Burkert	12	0.472	0.039	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC00891	VDII	5	3.263	0.653	1.000	60.000	0.799	N.A.	MAP nuisance fit
UGC00891	MOND	5	3.162	0.632	1.000	60.000	0.799	N.A.	MAP nuisance fit
UGC00891	NFW	5	2.102	0.420	1.000	60.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC00891	Burkert	5	0.316	0.063	1.000	60.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC01230	VDII	10	3.805	0.380	1.000	22.000	1.000	N.A.	MAP nuisance fit
UGC01230	MOND	10	3.020	0.302	1.000	22.000	1.252	N.A.	MAP nuisance fit
UGC01230	NFW	10	1.522	0.152	1.000	22.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC01230	Burkert	10	0.661	0.066	1.000	22.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC01281	VDII	20	5.774	0.289	1.000	42.000	1.000	N.A.	MAP nuisance fit
UGC01281	MOND	20	5.741	0.287	1.000	42.000	1.000	N.A.	MAP nuisance fit
UGC01281	NFW	20	5.676	0.284	1.000	42.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC01281	Burkert	20	0.991	0.050	1.000	42.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC02023	VDII	4	3.156	0.789	1.000	18.250	0.799	N.A.	MAP nuisance fit
UGC02023	MOND	4	2.932	0.733	1.000	18.250	0.799	N.A.	MAP nuisance fit
UGC02023	NFW	4	1.279	0.320	1.000	19.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC02023	Burkert	4	0.757	0.189	1.000	19.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC02259	VDII	8	1.805	0.226	1.047	41.000	1.568	N.A.	MAP nuisance fit
UGC02259	MOND	8	1.969	0.246	1.047	41.000	1.568	N.A.	MAP nuisance fit
UGC02259	NFW	8	0.116	0.015	1.000	41.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC02259	Burkert	8	0.466	0.058	1.000	41.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC02455	VDII	8	19.556	2.444	0.539	74.438	0.407	N.A.	MAP nuisance fit
UGC02455	MOND	8	19.519	2.440	0.539	74.438	0.407	N.A.	MAP nuisance fit
UGC02455	NFW	8	8.946	1.118	0.539	81.233	0.407	N.A.	MAP nuisance + 2-parameter halo
UGC02455	Burkert	8	12.655	1.582	0.693	77.350	0.407	N.A.	MAP nuisance + 2-parameter halo
UGC02487	VDII	17	0.732	0.043	1.220	36.000	1.000	N.A.	MAP nuisance fit
UGC02487	MOND	17	1.802	0.106	1.110	36.750	1.000	N.A.	MAP nuisance fit
UGC02487	NFW	17	0.234	0.014	1.000	36.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC02487	Burkert	17	0.338	0.020	1.000	36.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC02885	VDII	19	1.791	0.094	1.000	61.000	1.000	N.A.	MAP nuisance fit
UGC02885	MOND	19	2.271	0.120	1.000	61.000	1.000	N.A.	MAP nuisance fit
UGC02885	NFW	19	1.321	0.070	1.000	61.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC02885	Burkert	19	0.766	0.040	1.000	61.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC02916	VDII	43	10.511	0.244	0.953	49.250	0.799	N.A.	MAP nuisance fit
UGC02916	MOND	43	10.153	0.236	0.906	49.250	0.799	N.A.	MAP nuisance fit
UGC02916	NFW	43	10.237	0.238	0.937	50.000	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC02916	Burkert	43	8.068	0.188	0.937	49.000	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC02953	VDII	115	44.680	0.389	0.983	47.250	1.000	N.A.	MAP nuisance fit
UGC02953	MOND	115	49.435	0.430	0.965	47.250	1.000	N.A.	MAP nuisance fit
UGC02953	NFW	115	26.562	0.231	0.953	46.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC02953	Burkert	115	24.833	0.216	0.977	46.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC03205	VDII	48	6.507	0.136	0.970	67.000	1.000	N.A.	MAP nuisance fit

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Table S7: **Per-galaxy matched-refit summary (continued).**

Galaxy	Model	$N_{\rm pts}$	χ^2	χ^2_ν	$s_D^{\rm MAP}$	$i^{\rm MAP}$	$\Upsilon_{\rm disc}^{\rm MAP}$	$\Upsilon_{\rm bul}^{\rm MAP}$	notes
UGC03205	MOND	48	7.768	0.162	0.970	67.000	1.000	N.A.	MAP nuisance fit
UGC03205	NFW	48	6.420	0.134	1.000	67.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC03205	Burkert	48	5.508	0.115	1.000	67.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC03546	VDII	30	6.785	0.226	1.000	55.000	0.638	N.A.	MAP nuisance fit
UGC03546	MOND	30	8.231	0.274	1.000	55.000	0.638	N.A.	MAP nuisance fit
UGC03546	NFW	30	1.927	0.064	1.000	55.000	0.549	N.A.	MAP nuisance + 2-parameter halo
UGC03546	Burkert	30	1.260	0.042	1.000	55.000	0.549	N.A.	MAP nuisance + 2-parameter halo
UGC03580	VDII	45	22.275	0.495	0.956	63.000	0.407	N.A.	MAP nuisance fit
UGC03580	MOND	45	25.373	0.564	0.956	63.000	0.407	N.A.	MAP nuisance fit
UGC03580	NFW	45	10.686	0.237	0.941	63.000	0.407	N.A.	MAP nuisance + 2-parameter halo
UGC03580	Burkert	45	6.853	0.152	0.941	63.000	0.407	N.A.	MAP nuisance + 2-parameter halo
UGC04278	VDII	23	8.863	0.385	1.000	51.000	1.000	N.A.	MAP nuisance fit
UGC04278	MOND	23	9.043	0.393	1.000	51.000	1.000	N.A.	MAP nuisance fit
UGC04278	NFW	23	6.244	0.271	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC04278	Burkert	23	1.438	0.063	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC04305	VDII	21	16.124	0.768	0.550	63.000	0.638	N.A.	MAP nuisance fit
UGC04305	MOND	21	17.441	0.831	0.550	66.000	0.638	N.A.	MAP nuisance fit
UGC04305	NFW	21	6.313	0.301	0.850	68.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC04305	Burkert	21	5.760	0.274	0.850	72.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC04325	VDII	8	2.630	0.329	1.000	81.388	1.568	N.A.	MAP nuisance fit
UGC04325	MOND	8	2.123	0.265	1.046	81.388	1.568	N.A.	MAP nuisance fit
UGC04325	NFW	8	0.791	0.099	1.000	82.233	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC04325	Burkert	8	0.159	0.020	1.000	82.233	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC04483	VDII	7	1.301	0.186	0.915	11.100	1.000	N.A.	MAP nuisance fit
UGC04483	MOND	7	1.070	0.153	1.000	11.100	0.799	N.A.	MAP nuisance fit
UGC04483	NFW	7	1.139	0.163	1.000	13.133	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC04483	Burkert	7	0.540	0.077	1.000	13.133	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC04499	VDII	9	2.698	0.300	1.000	47.250	0.799	N.A.	MAP nuisance fit
UGC04499	MOND	9	3.281	0.365	1.000	51.000	0.799	N.A.	MAP nuisance fit
UGC04499	NFW	9	2.839	0.315	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC04499	Burkert	9	1.410	0.157	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05005	VDII	8	1.199	0.150	0.901	41.000	0.799	N.A.	MAP nuisance fit
UGC05005	MOND	8	1.733	0.217	1.000	41.000	0.799	N.A.	MAP nuisance fit
UGC05005	NFW	8	0.621	0.078	1.000	41.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05005	Burkert	8	0.100	0.012	1.000	41.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05253	VDII	73	8.524	0.117	0.969	21.000	0.799	N.A.	MAP nuisance fit
UGC05253	MOND	73	12.638	0.173	1.000	18.225	1.000	N.A.	MAP nuisance fit
UGC05253	NFW	73	7.374	0.101	1.000	21.000	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC05253	Burkert	73	5.258	0.072	1.000	21.000	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC05414	VDII	6	1.317	0.219	1.000	17.188	1.000	N.A.	MAP nuisance fit
UGC05414	MOND	6	1.388	0.231	1.000	17.188	1.000	N.A.	MAP nuisance fit
UGC05414	NFW	6	1.301	0.217	1.000	21.250	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05414	Burkert	6	0.090	0.015	1.000	21.250	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05716	VDII	12	0.808	0.067	1.107	54.000	1.000	N.A.	MAP nuisance fit
UGC05716	MOND	12	0.371	0.031	1.107	54.000	1.252	N.A.	MAP nuisance fit
UGC05716	NFW	12	0.136	0.011	1.000	54.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05716	Burkert	12	0.947	0.079	1.000	54.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05721	VDII	23	10.821	0.470	1.018	64.725	1.000	N.A.	MAP nuisance fit
UGC05721	MOND	23	11.991	0.521	1.018	64.725	1.000	N.A.	MAP nuisance fit
UGC05721	NFW	23	4.225	0.184	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05721	Burkert	23	1.040	0.045	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05750	VDII	8	1.123	0.140	0.939	64.000	0.799	N.A.	MAP nuisance fit
UGC05750	MOND	8	0.854	0.107	0.939	64.000	0.799	N.A.	MAP nuisance fit
UGC05750	NFW	8	0.514	0.064	1.000	64.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05750	Burkert	8	0.119	0.015	1.000	64.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05764	VDII	10	4.925	0.493	1.051	50.000	1.252	N.A.	MAP nuisance fit
UGC05764	MOND	10	5.484	0.548	1.051	50.000	1.252	N.A.	MAP nuisance fit
UGC05764	NFW	10	1.568	0.157	1.000	41.000	1.000	N.A.	MAP nuisance + 2-parameter halo

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Table S7: **Per-galaxy matched-refit summary (continued).**

Galaxy	Model	$N_{\rm pts}$	χ^2	χ^2_ν	$s_D^{\rm MAP}$	$i^{\rm MAP}$	$\Upsilon_{\rm disc}^{\rm MAP}$	$\Upsilon_{\rm bul}^{\rm MAP}$	notes
UGC05764	Burkert	10	0.601	0.060	1.000	41.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05829	VDII	10	0.402	0.040	1.000	34.000	1.000	N.A.	MAP nuisance fit
UGC05829	MOND	10	0.284	0.028	1.000	34.000	1.000	N.A.	MAP nuisance fit
UGC05829	NFW	10	0.044	0.004	1.000	34.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05829	Burkert	10	0.374	0.037	1.000	34.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05918	VDII	8	0.795	0.099	1.000	46.000	1.000	N.A.	MAP nuisance fit
UGC05918	MOND	8	0.908	0.113	1.000	46.000	1.000	N.A.	MAP nuisance fit
UGC05918	NFW	8	0.300	0.038	1.000	46.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05918	Burkert	8	0.067	0.008	1.000	46.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05986	VDII	15	8.146	0.543	1.048	89.000	1.000	N.A.	MAP nuisance fit
UGC05986	MOND	15	8.583	0.572	1.048	89.000	1.000	N.A.	MAP nuisance fit
UGC05986	NFW	15	5.300	0.353	1.000	89.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05986	Burkert	15	1.509	0.101	1.000	89.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05999	VDII	5	1.923	0.385	1.000	22.000	0.799	N.A.	MAP nuisance fit
UGC05999	MOND	5	1.904	0.381	1.000	22.000	0.799	N.A.	MAP nuisance fit
UGC05999	NFW	5	1.510	0.302	1.000	22.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC05999	Burkert	5	0.661	0.132	1.000	22.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06399	VDII	9	1.417	0.157	1.000	75.000	1.000	N.A.	MAP nuisance fit
UGC06399	MOND	9	1.562	0.174	1.000	75.000	1.000	N.A.	MAP nuisance fit
UGC06399	NFW	9	1.138	0.126	1.000	75.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06399	Burkert	9	0.133	0.015	1.000	75.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06446	VDII	17	2.083	0.123	1.032	51.000	1.568	N.A.	MAP nuisance fit
UGC06446	MOND	17	2.544	0.150	1.032	51.000	1.568	N.A.	MAP nuisance fit
UGC06446	NFW	17	0.817	0.048	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06446	Burkert	17	0.682	0.040	1.000	51.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06614	VDII	13	3.770	0.290	0.940	71.000	0.799	N.A.	MAP nuisance fit
UGC06614	MOND	13	2.530	0.195	1.000	71.000	0.638	N.A.	MAP nuisance fit
UGC06614	NFW	13	1.774	0.136	1.000	71.000	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC06614	Burkert	13	0.838	0.064	1.000	71.000	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC06628	VDII	7	7.721	1.103	0.830	26.500	0.638	N.A.	MAP nuisance fit
UGC06628	MOND	7	7.057	1.008	0.830	26.500	0.638	N.A.	MAP nuisance fit
UGC06628	NFW	7	1.260	0.180	0.886	29.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06628	Burkert	7	1.714	0.245	0.886	31.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06667	VDII	9	6.801	0.756	1.167	89.000	1.964	N.A.	MAP nuisance fit
UGC06667	MOND	9	7.627	0.847	1.167	89.000	1.964	N.A.	MAP nuisance fit
UGC06667	NFW	9	1.091	0.121	1.000	89.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06667	Burkert	9	0.064	0.007	1.000	89.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06786	VDII	45	9.809	0.218	1.079	22.600	0.799	N.A.	MAP nuisance fit
UGC06786	MOND	45	11.284	0.251	1.079	22.600	0.799	N.A.	MAP nuisance fit
UGC06786	NFW	45	2.853	0.063	1.000	22.600	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC06786	Burkert	45	2.104	0.047	1.000	22.600	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC06787	VDII	71	41.624	0.586	0.896	22.900	0.799	N.A.	MAP nuisance fit
UGC06787	MOND	71	44.257	0.623	0.896	22.900	0.799	N.A.	MAP nuisance fit
UGC06787	NFW	71	10.600	0.149	0.862	22.900	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC06787	Burkert	71	8.882	0.125	0.862	22.900	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC06818	VDII	8	9.142	1.143	1.000	75.000	0.509	N.A.	MAP nuisance fit
UGC06818	MOND	8	8.844	1.106	1.000	75.000	0.509	N.A.	MAP nuisance fit
UGC06818	NFW	8	7.877	0.985	1.000	75.000	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC06818	Burkert	8	2.944	0.368	1.000	75.000	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC06917	VDII	11	1.045	0.095	1.000	56.000	1.000	N.A.	MAP nuisance fit
UGC06917	MOND	11	1.159	0.105	1.000	56.000	1.000	N.A.	MAP nuisance fit
UGC06917	NFW	11	0.728	0.066	1.000	56.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06917	Burkert	11	0.299	0.027	1.000	56.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06923	VDII	6	2.066	0.344	0.917	65.000	0.799	N.A.	MAP nuisance fit
UGC06923	MOND	6	2.153	0.359	0.917	65.000	0.799	N.A.	MAP nuisance fit
UGC06923	NFW	6	2.852	0.475	1.000	65.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06923	Burkert	6	1.785	0.298	1.000	65.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06930	VDII	10	0.771	0.077	0.917	32.000	1.000	N.A.	MAP nuisance fit

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Table S7: **Per-galaxy matched-refit summary (continued).**

Galaxy	Model	$N_{\rm pts}$	χ^2	χ^2_ν	$s_D^{\rm MAP}$	$i^{\rm MAP}$	$\Upsilon_{\rm disc}^{\rm MAP}$	$\Upsilon_{\rm bul}^{\rm MAP}$	notes
UGC06930	MOND	10	1.174	0.117	1.000	32.000	1.000	N.A.	MAP nuisance fit
UGC06930	NFW	10	0.745	0.074	1.000	32.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06930	Burkert	10	0.400	0.040	1.000	32.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06973	VDII	9	7.772	0.864	0.931	68.000	0.509	N.A.	MAP nuisance fit
UGC06973	MOND	9	8.469	0.941	0.931	68.000	0.509	N.A.	MAP nuisance fit
UGC06973	NFW	9	8.444	0.938	0.908	67.000	0.549	N.A.	MAP nuisance + 2-parameter halo
UGC06973	Burkert	9	2.513	0.279	0.908	71.000	0.407	N.A.	MAP nuisance + 2-parameter halo
UGC06983	VDII	17	2.780	0.164	1.083	49.000	1.000	N.A.	MAP nuisance fit
UGC06983	MOND	17	3.134	0.184	1.083	49.000	1.000	N.A.	MAP nuisance fit
UGC06983	NFW	17	2.098	0.123	1.000	49.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC06983	Burkert	17	1.063	0.063	1.000	49.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07089	VDII	12	2.552	0.213	0.917	78.500	0.638	N.A.	MAP nuisance fit
UGC07089	MOND	12	2.789	0.232	0.833	78.500	0.799	N.A.	MAP nuisance fit
UGC07089	NFW	12	2.087	0.174	1.000	78.500	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07089	Burkert	12	0.278	0.023	1.000	78.500	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07125	VDII	13	16.216	1.247	0.939	89.000	0.509	N.A.	MAP nuisance fit
UGC07125	MOND	13	13.877	1.067	0.939	89.000	0.509	N.A.	MAP nuisance fit
UGC07125	NFW	13	1.588	0.122	1.000	89.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07125	Burkert	13	0.712	0.055	1.000	89.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07151	VDII	11	2.041	0.186	0.901	42.000	1.000	N.A.	MAP nuisance fit
UGC07151	MOND	11	1.985	0.180	0.901	42.000	1.000	N.A.	MAP nuisance fit
UGC07151	NFW	11	1.644	0.149	1.000	42.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07151	Burkert	11	0.649	0.059	1.000	42.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07232	VDII	4	1.827	0.457	0.875	71.650	1.000	N.A.	MAP nuisance fit
UGC07232	MOND	4	1.859	0.465	0.875	71.650	1.000	N.A.	MAP nuisance fit
UGC07232	NFW	4	1.786	0.447	1.000	71.650	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07232	Burkert	4	0.577	0.144	1.000	71.650	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07261	VDII	7	0.674	0.096	1.000	31.000	1.000	N.A.	MAP nuisance fit
UGC07261	MOND	7	0.682	0.097	1.000	31.000	1.000	N.A.	MAP nuisance fit
UGC07261	NFW	7	0.345	0.049	1.000	31.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07261	Burkert	7	0.785	0.112	1.000	31.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07323	VDII	10	1.668	0.167	1.000	47.000	0.638	N.A.	MAP nuisance fit
UGC07323	MOND	10	1.465	0.146	1.000	47.000	0.638	N.A.	MAP nuisance fit
UGC07323	NFW	10	1.402	0.140	1.000	47.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07323	Burkert	10	0.331	0.033	1.000	47.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07399	VDII	10	3.552	0.355	1.037	39.250	1.252	N.A.	MAP nuisance fit
UGC07399	MOND	10	2.789	0.279	1.037	43.375	1.000	N.A.	MAP nuisance fit
UGC07399	NFW	10	0.482	0.048	1.000	31.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07399	Burkert	10	0.176	0.018	1.000	31.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07524	VDII	31	5.488	0.177	1.000	42.000	0.799	N.A.	MAP nuisance fit
UGC07524	MOND	31	5.876	0.190	1.000	42.000	1.000	N.A.	MAP nuisance fit
UGC07524	NFW	31	4.363	0.141	1.000	42.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07524	Burkert	31	0.842	0.027	1.000	42.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07559	VDII	6	1.405	0.234	0.625	52.000	1.000	N.A.	MAP nuisance fit
UGC07559	MOND	6	1.261	0.210	0.625	52.000	1.000	N.A.	MAP nuisance fit
UGC07559	NFW	6	1.044	0.174	1.000	52.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07559	Burkert	6	0.369	0.061	1.000	52.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07577	VDII	5	0.923	0.185	0.625	22.550	0.799	N.A.	MAP nuisance fit
UGC07577	MOND	5	1.342	0.268	0.625	22.550	1.000	N.A.	MAP nuisance fit
UGC07577	NFW	5	0.889	0.178	0.833	32.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07577	Burkert	5	0.993	0.199	1.000	32.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07603	VDII	12	6.307	0.526	1.000	12.350	0.799	N.A.	MAP nuisance fit
UGC07603	MOND	12	6.680	0.557	1.000	12.350	0.799	N.A.	MAP nuisance fit
UGC07603	NFW	12	2.689	0.224	1.000	9.900	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07603	Burkert	12	0.507	0.042	1.000	9.900	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07608	VDII	7	4.218	0.603	1.000	62.875	1.252	N.A.	MAP nuisance fit
UGC07608	MOND	7	4.753	0.679	1.000	62.875	1.252	N.A.	MAP nuisance fit
UGC07608	NFW	7	0.194	0.028	1.000	61.000	1.000	N.A.	MAP nuisance + 2-parameter halo

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Table S7: **Per-galaxy matched-refit summary (continued).**

Galaxy	Model	$N_{\rm pts}$	χ^2	χ^2_ν	$s_D^{\rm MAP}$	$i^{\rm MAP}$	$\Upsilon_{\rm disc}^{\rm MAP}$	$\Upsilon_{\rm bul}^{\rm MAP}$	notes
UGC07608	Burkert	7	0.060	0.009	1.000	61.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07690	VDII	7	0.378	0.054	0.895	31.000	1.000	N.A.	MAP nuisance fit
UGC07690	MOND	7	0.353	0.050	0.895	31.000	1.000	N.A.	MAP nuisance fit
UGC07690	NFW	7	0.372	0.053	1.000	31.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07690	Burkert	7	0.203	0.029	1.000	31.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07866	VDII	6	0.407	0.068	0.893	28.700	1.000	N.A.	MAP nuisance fit
UGC07866	MOND	6	0.266	0.044	0.893	28.700	1.000	N.A.	MAP nuisance fit
UGC07866	NFW	6	0.044	0.007	1.000	32.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC07866	Burkert	6	0.130	0.022	1.000	32.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC08286	VDII	17	3.007	0.177	1.057	13.500	1.000	N.A.	MAP nuisance fit
UGC08286	MOND	17	3.284	0.193	1.057	13.500	1.000	N.A.	MAP nuisance fit
UGC08286	NFW	17	4.492	0.264	1.000	10.667	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC08286	Burkert	17	0.932	0.055	1.000	10.667	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC08490	VDII	30	1.523	0.051	1.315	32.000	1.000	N.A.	MAP nuisance fit
UGC08490	MOND	30	2.254	0.075	1.315	32.000	1.000	N.A.	MAP nuisance fit
UGC08490	NFW	30	0.442	0.015	1.000	32.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC08490	Burkert	30	0.970	0.032	1.000	32.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC08550	VDII	11	1.427	0.130	1.000	89.000	1.252	N.A.	MAP nuisance fit
UGC08550	MOND	11	1.181	0.107	1.069	89.000	1.252	N.A.	MAP nuisance fit
UGC08550	NFW	11	0.421	0.038	1.000	89.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC08550	Burkert	11	0.361	0.033	1.000	89.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC08699	VDII	41	11.386	0.278	1.000	19.212	1.000	N.A.	MAP nuisance fit
UGC08699	MOND	41	11.964	0.292	1.000	19.212	1.000	N.A.	MAP nuisance fit
UGC08699	NFW	41	6.293	0.153	0.871	23.950	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC08699	Burkert	41	6.049	0.148	0.871	17.633	1.350	N.A.	MAP nuisance + 2-parameter halo
UGC08837	VDII	8	9.657	1.207	1.000	38.000	0.799	N.A.	MAP nuisance fit
UGC08837	MOND	8	9.356	1.169	1.000	38.000	0.799	N.A.	MAP nuisance fit
UGC08837	NFW	8	8.980	1.123	1.000	62.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC08837	Burkert	8	4.206	0.526	1.000	54.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC09037	VDII	22	12.105	0.550	0.866	65.000	0.509	N.A.	MAP nuisance fit
UGC09037	MOND	22	12.972	0.590	0.866	65.000	0.509	N.A.	MAP nuisance fit
UGC09037	NFW	22	13.646	0.620	0.911	65.000	0.549	N.A.	MAP nuisance + 2-parameter halo
UGC09037	Burkert	22	4.604	0.209	0.911	65.000	0.407	N.A.	MAP nuisance + 2-parameter halo
UGC09133	VDII	68	6.782	0.100	0.970	53.000	0.799	N.A.	MAP nuisance fit
UGC09133	MOND	68	8.781	0.129	1.000	52.250	0.799	N.A.	MAP nuisance fit
UGC09133	NFW	68	5.761	0.085	0.961	53.000	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC09133	Burkert	68	6.309	0.093	1.000	52.000	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC09992	VDII	5	1.289	0.258	1.000	8.750	1.000	N.A.	MAP nuisance fit
UGC09992	MOND	5	1.099	0.220	1.000	8.750	1.000	N.A.	MAP nuisance fit
UGC09992	NFW	5	0.092	0.018	1.000	10.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC09992	Burkert	5	0.041	0.008	1.000	10.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC10310	VDII	7	0.547	0.078	1.000	34.000	1.000	N.A.	MAP nuisance fit
UGC10310	MOND	7	0.450	0.064	1.000	34.000	1.000	N.A.	MAP nuisance fit
UGC10310	NFW	7	0.341	0.049	1.000	34.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC10310	Burkert	7	0.055	0.008	1.000	34.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC11455	VDII	36	21.418	0.595	0.934	89.000	0.638	N.A.	MAP nuisance fit
UGC11455	MOND	36	21.184	0.588	0.867	89.000	0.638	N.A.	MAP nuisance fit
UGC11455	NFW	36	22.967	0.638	0.823	89.000	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC11455	Burkert	36	14.980	0.416	0.823	89.000	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC11557	VDII	10	6.786	0.679	0.872	46.500	0.509	N.A.	MAP nuisance fit
UGC11557	MOND	10	6.445	0.645	0.872	46.500	0.509	N.A.	MAP nuisance fit
UGC11557	NFW	10	4.562	0.456	0.915	48.000	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC11557	Burkert	10	4.463	0.446	0.915	48.000	0.741	N.A.	MAP nuisance + 2-parameter halo
UGC11820	VDII	10	1.065	0.107	1.000	31.000	1.000	N.A.	MAP nuisance fit
UGC11820	MOND	10	1.106	0.111	1.000	31.000	1.000	N.A.	MAP nuisance fit
UGC11820	NFW	10	1.250	0.125	1.000	31.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC11820	Burkert	10	2.640	0.264	1.000	31.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC11914	VDII	65	3.578	0.055	0.969	30.250	1.000	N.A.	MAP nuisance fit

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Table S7: **Per-galaxy matched-refit summary (continued).**

Galaxy	Model	$N_{\rm pts}$	χ^2	χ^2_ν	$s_D^{\rm MAP}$	$i^{\rm MAP}$	$\Upsilon_{\rm disc}^{\rm MAP}$	$\Upsilon_{\rm bul}^{\rm MAP}$	notes
UGC11914	MOND	65	3.512	0.054	0.937	30.250	1.000	N.A.	MAP nuisance fit
UGC11914	NFW	65	2.753	0.042	0.958	30.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC11914	Burkert	65	2.815	0.043	0.958	30.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC12506	VDII	31	7.649	0.247	1.074	86.000	1.252	N.A.	MAP nuisance fit
UGC12506	MOND	31	7.206	0.232	1.074	86.000	1.252	N.A.	MAP nuisance fit
UGC12506	NFW	31	1.487	0.048	1.000	86.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC12506	Burkert	31	5.346	0.172	1.000	86.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC12632	VDII	15	1.368	0.091	1.000	31.000	1.000	N.A.	MAP nuisance fit
UGC12632	MOND	15	0.981	0.065	1.000	31.000	1.252	N.A.	MAP nuisance fit
UGC12632	NFW	15	0.663	0.044	1.000	31.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC12632	Burkert	15	0.168	0.011	1.000	31.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC12732	VDII	16	0.561	0.035	1.000	39.000	1.252	N.A.	MAP nuisance fit
UGC12732	MOND	16	0.930	0.058	1.000	39.000	1.252	N.A.	MAP nuisance fit
UGC12732	NFW	16	0.148	0.009	1.000	39.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGC12732	Burkert	16	1.018	0.064	1.000	39.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGCA281	VDII	6	0.636	0.106	1.000	82.225	0.799	N.A.	MAP nuisance fit
UGCA281	MOND	6	0.687	0.115	0.910	82.225	1.000	N.A.	MAP nuisance fit
UGCA281	NFW	6	0.210	0.035	1.000	79.967	1.000	N.A.	MAP nuisance + 2-parameter halo
UGCA281	Burkert	6	0.027	0.004	1.000	79.967	1.000	N.A.	MAP nuisance + 2-parameter halo
UGCA442	VDII	8	1.431	0.179	1.000	23.100	1.000	N.A.	MAP nuisance fit
UGCA442	MOND	8	1.631	0.204	1.000	23.100	1.000	N.A.	MAP nuisance fit
UGCA442	NFW	8	0.671	0.084	1.000	23.100	1.000	N.A.	MAP nuisance + 2-parameter halo
UGCA442	Burkert	8	0.680	0.085	1.000	23.100	1.000	N.A.	MAP nuisance + 2-parameter halo
UGCA444	VDII	35	2.402	0.069	1.000	52.000	1.000	N.A.	MAP nuisance fit
UGCA444	MOND	35	2.714	0.078	1.000	52.000	1.000	N.A.	MAP nuisance fit
UGCA444	NFW	35	0.775	0.022	1.000	52.000	1.000	N.A.	MAP nuisance + 2-parameter halo
UGCA444	Burkert	35	2.469	0.071	1.000	52.000	1.000	N.A.	MAP nuisance + 2-parameter halo

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